



Parallel Kinematic Delta Robot Pen Plotter

BEng PROJECT FINAL DISSERTATION

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Summary

This report summarises a bachelor's individual project that includes the design, manufacturing, and programming of a parallel kinematic robot, specifically a delta robot, functioning as a pen plotter for icing to be used for cake decoration. This device was designed with the intention to be used as an engaging educational demonstration of parallel kinematic robotics.

During this project the following has been achieved: A full SolidWorks model of the mechanical system has been completed, including dimensions and materials. The 3D printed parts modelled in CAD have been printed, and the machined elements have been manufactured. The system has been assembled, and mechanical parts tested for carrying out functions. The electronic components of the system were selected, and the circuit board was designed and assembled with the servo motors. The robotic control program for the servos has been coded and downloaded onto the robot's microcontroller, and the system has been successfully demonstrated in two motor control experiments.

During the prototyping phase, multiple alterations were made to the schematics. One change was that the parts that connect the bottom of the robot's arms to the motors were changed to slot over the motor's included plastic arms rather than slotting over the motor head itself, since the Nylon 12 material used for the parts did not have the precision necessary to create the appropriate pattern to slot into the motor head's teeth.

Potential further work for this project could include a more thorough servo control program, with a full decoration demonstration. The icing ejection system could be designed and implemented into the model. Had there been more time, these would have been part of the project itself.

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ACRONYMS

Acronym	Meaning
'V.'	Version
'Pi'	Raspberry Pi
'DOF'	Degrees of Freedom

1. INTRODUCTION.

1.1. Background and Context

The use of robotics in practical demonstrations for machine kinematics is commonplace in technological schools and courses. However, it is widely believed by many educational researchers that successful teaching involves making the content engaging. Leonardo Da Vinci, the inventor of one of the earliest recorded humanoid automatons in western history (the 'mechanical knight', constructed in 1495 approximately), once said "Study without desire spoils the memory, and it retains nothing that it takes in" [1]. This project aims to engage students in the demonstration by relating the use of parallel kinematic robotics to cake decoration, in the hopes that it will spark inspiration in those with and without prior interest in robotics.

Parallel kinematic robots have become a valuable member of the robotics family in recent years due to globalisation and the emergence of the automation industry. Despite their presence in many factories, including for fruit production lines, parallel kinematic robotics are not commonly taught about in technological educational programmes. This project proposes an engaging educational example for parallel kinematic robotics. It will demonstrate the possibilities for parallel kinematic robots to carry out practical activity on topics of interest, while staying small and light enough to be easily carried and transported for academic demonstrations.

In order to determine how to approach the design of the delta robot, it is first necessary to understand what a delta robot is and how it works. A delta robot is a type of parallel kinematic robotic system that consists of at least two arms connected to a base. Usually, the arms are moved by actuation from linear or rotary motors. Unlike traditional serial robots where the end-effector (the targeted moving part) is attached to the last link of the arm, in delta robots, the end-effector is directly attached to the moving platform that is suspended by the arms. This configuration offers some advantages, such as higher speeds, accelerations, and precision, as well as a larger workspace compared to serial robots.

Delta robots are commonly used in high-speed pick-and-place operations, for example the food and beverage industry, where they can handle items quickly and accurately. Other examples of delta robot use are in 3D printing, assembly, and inspection. These are all operations where precision and speed are crucial.

Since most examples of industrial delta robots come from the food packing and processing industry and the project aimed to create an engaging and interesting presentation, the project instead took inspiration from the commercial use of delta robots, as these would be likely to serve novelty purposes more apt for an engaging educational demonstration. The most popular use of commercial delta robots is in 3D printing delta robots, which carry out tasks more similar to those of a pen plotter than a packaging robot. This led to the idea of a cupcake decorating robot: a delta robot pen plotter which can eject icing at different programmed positions on a cupcake.

2. LITERATURE REVIEW

During this project's literature review, there have been a wide number of sources that have provided inspiration and context for the project. However, only the most relevant ones to the inspiration and design processes have been included in this report, in order to allow for thorough discussion of those included.

Ilian Bonev discusses the history behind Delta robots in 'Delta Parallel Robot — the Story of Success' [2], where he tells us about Raymond Clavel's (professor at EPFL - École Polytechnique Fédérale de Lausanne) work in the early 1980s, and the unlikely success of his innovation. The initial inspiration for the Delta robot comes from the unique shape of the parallelogram. Bonev presents a concise explanation of the key components of a Delta robot:

"The basic idea behind the Delta robot design is the use of parallelograms. A parallelogram allows an output link to remain at a fixed orientation with respect to an input link. The use of three such parallelograms restrain completely the orientation of the mobile platform which remains only with three purely translational degrees of freedom." - Ilian Bonev, Delta Parallel Robot — the Story of Success [2]

This research lays the groundwork for important factors to be considered in this project's design and inspired the use of a Delta robot as an effective demonstration of parallel Kinematic Robotics given the simple concept of being able to translate the mobile platform with 'three purely translational degrees of freedom', and the clear applications for Delta robots having practical use as pen plotters. Using a robot with the clear structure and function of a Delta Robot as a pen plotter allows the students of the educational demonstration to see the real-life applications of parallel kinematic robots as the Delta robot can complete a task that students will be familiar with and able to relate to, rather than just witnessing machinery move with no clear intention.

Figure 1 shows the simple structure of a basic Delta robot and provides a clear example of how the robot components should be assembled.

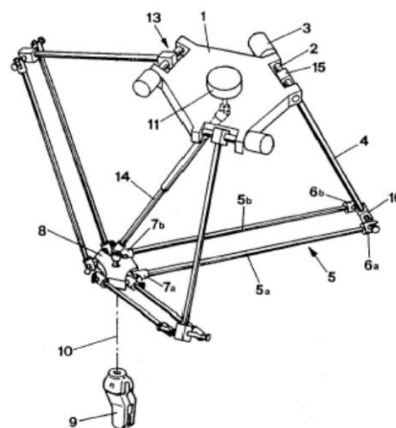


Figure 1- Schematic of the Delta robot (from US patent No. 4,976,582)

This design for the original Delta robot differs from this project's design since this project aims to adapt the original Delta robot design to allow its use as a cake decorating robot, and therefore an icing release syringe will need to be incorporated into the design, and this project will adapt the design to have a more compact and portable form factor. Also, where the original Delta robots were made mainly from metal, this project aims to use mainly PLA 3d printing filament and wood. However, Bonev outlines a clear criterion of what defines a Delta robot, which has provided a good basis for the overall structure of the robot that this project aims to create.

In 'Educational Robots and Computational Thinking' [3], Dave Catlin and Dr John Woollard discuss how "thinking processes and disciplines used by computer scientists would benefit students of all subjects" [3]. The paper explores the idea for use of educational robots as a means of engaging students in computational thinking (or 'CT') and discusses the psychological relevance of robotics and engaged learning to early childhood memories of robotics. Catlin and Woollard discuss how this early fascination helped to inspire the scientific interest in both the authors and their research subjects alike. This provides support for this project's intention to use a robotic device to explain the theory behind parallel kinematics and helped inspire the idea for incorporating a whimsical function of cake decoration to appeal to the student with the same child-like wonder that initially sparked interest in the authors during their childhoods.

The initial inspiration for the use of an Arduino to control the system came from Sheetal N. Patil and Prashant G. Patil's 'Implementation of Arduino UNO based Two Directional [2D] Plotter' [4]. This paper outlines the research into which functions of different micro-processors are most useful for different components used in their pen plotter, such as a servo motor and a Bluetooth module.

This paper explores Arduino application for two-dimensional plotters. While this project will be using a three-dimensional plotter, much of the research in this paper highlights some of the factors that would need to be considered in the control of this project's design as well.

After research on Arduino use in robotics systems, the importance of exploring alternative microcontrollers was highlighted. It was through Joshua Hrisco's 'Raspberry Pi Camera Panning with a Servo Motor' [5] that the advantage of Raspberry Pi over Arduino in the context of robotics became clear. Hrisco outlines how "The general-purpose inputs/outputs are beneficial for learning about sensors, motors, and indicators - all without the need for external data acquisition cards or control devices". The Raspberry Pi's ability to function independently from a controller board is a great advantage in a system which aims to be as compact and self-sufficient as possible, since it allows for the use of control electronics within the mechanical body of the robot without requiring the physical space for an external controller board and the additional circuitry that would be necessary alongside it.

The design of a delta robot also requires some mathematical analysis in the form of kinematic modelling. In approaching this aspect of the project, the 'Delta Robot Kinematics' [6] tutorial on the 'Trossen Robotics Community' website aided in the mathematical method used in the reverse kinematic model. The tutorial explains "If we want to build our own delta robot, we need to solve two problems. First, if we know the desired position of the end effector (for example, we want to catch pancake in the point with coordinates X, Y, Z), we need to determine the corresponding angles of each of three arms (joint angles) to set motors (and, thereby, the end effector) in proper position for picking. The process of such determining is known as inverse kinematics. And, in the second place, if we know joint angles (for example, we've read the values of motor encoders), we need to determine the position of the end effector (e.g., to make some corrections of its current position). This is forward kinematics problem." This case study demonstrates the reverse kinematic model calculations, and this method is used in the project's own kinematic model which appears in section 3.4: Reverse and Forward Kinematic Model.

In understanding the further applications of parallel kinematics in delta robots, Robert L. Williams II's 'The Delta Parallel Robot: Kinematics Solutions' [7] provided exposition on the kinematic model of one of the most successful commercial robots to date. The robot analysed in this paper has 4 DOF, as well as a fourth inner leg to control a single rotational freedom at the end-effector platform. However, the kinematic analysis is of a 3 DOF system since that is widely applied by most modern delta robots. Williams explains that "As often happens, the Kutzbach equation fails because the result must obviously be 3-dof. This result predicts the Delta is a severely overconstrained statically indeterminate structure, which is incorrect. The Kutzbach equation knows nothing about special geometry – in the Delta Robot case, there are three parallel four-bar mechanisms. The overall robot would work kinematically identically to the original Delta Robot if we removed one of the long parallel four-bar mechanism links, along with two revolute joints each." This more closely contextualises the kinematic analysis in this paper with the analysis necessary for this project's robot, which also has 3 DOF and therefore will need to follow a mathematically similar kinematic analysis.

3. Designing and Creating the Robotic System

3.1. Mechanical System Design

This section will outline the machine schematic and how the system itself works before exploring some of the major changes made between different model iterations.

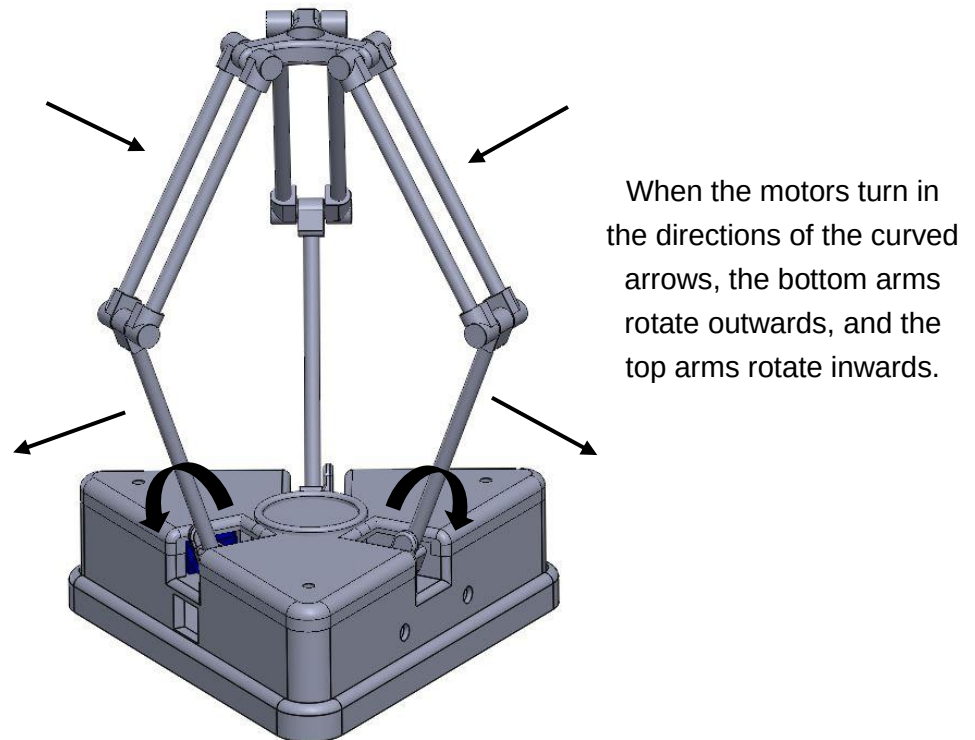
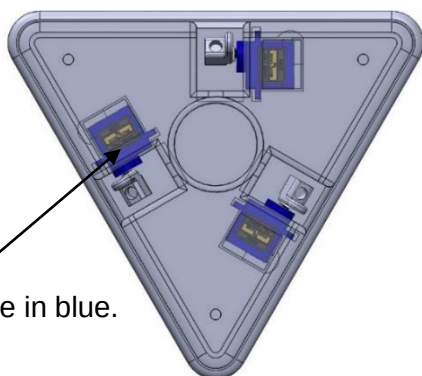


Figure 2 - V.5 Model

Figure 2 shows a side-on view of the final version model, where the overall assembly is made clear. The machine works by having three actuators mounted around a base, each of which is attached to an arm whose rotational movement is controlled by the actuator. Each single arm is attached at the top to another arm (consisting of two long rods instead of one) the top of which attaches to the moving platform which holds the nozzle, which in turn is attached to an icing feeding tube (nozzle and feeding tube are not pictured in model).



The base holds the three actuators in three separate slots where they can be fixed into place by a screw on each side of the actuator. The base lid is screwed onto the base and fits over the actuator slots, leaving gaps to allow for the single rod arms to rotate. Figure 3 pictures the base lid fitting over the actuators in their slots in the base.

Figure 3 – Top-Down view of Base with (transparent) Base Lid fitting over Actuators

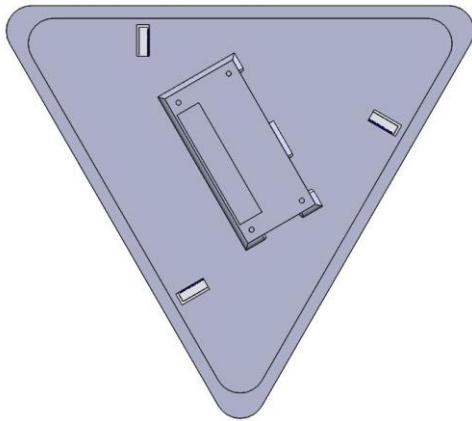


Figure 4 - Bottom-Up view of (transparent) Base Tray and Base with Raspberry Pi Mount

The base lid also features a circular zone within which the cupcake can be placed for decoration. The intention is that when running the cake decoration demonstration programme, the actuators will rotate the arms according to the kinematically analysed calculations, lowering the platform into position to carry out the movements to decorate the cake.

The electrical components, including the Raspberry Pi, will be concealed inside the base part, as there is a chamber underneath the base plate's actuator panel. The underside of the actuator panel has a mounting board for the Raspberry Pi, as is shown in Figure 4. There are also holes in the actuator panel in each actuator slot through which cables can be fed between the Raspberry Pi and the actuators.

The entire base part is open at the bottom to allow access to the electrical components inside, so in order for this to be concealed, the base slots into a base tray. Figure 5 depicts a blown-up view of the three parts which make up the base – the main 'base' part, the 'base lid' and the 'base tray'.

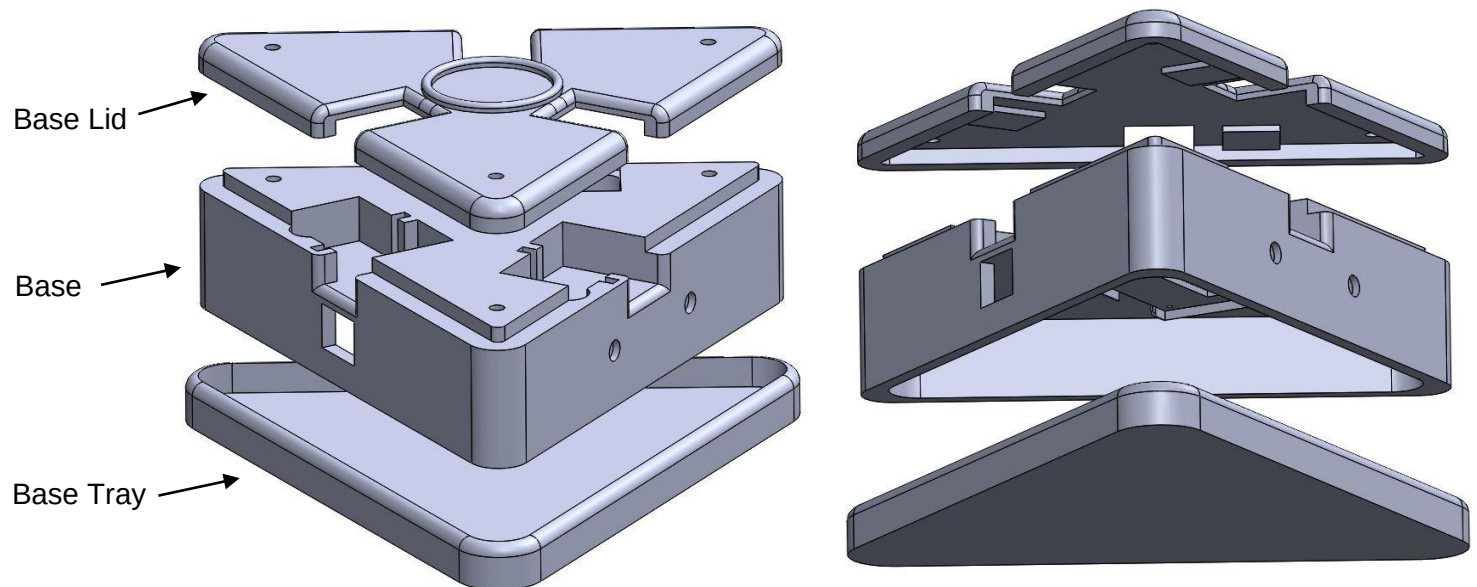


Figure 5 - Side by side views of V.5's blown-up base parts from above and below

During the design process, as in every project, there were many obstacles and puzzles to solve. There ended up being 5 different major iterations of the design. Although in this report the manufacturing phase is not covered until later, in reality a lot of the major design changes occurred during the prototyping phase where the manufactured parts had to be tested. The following section will outline some of the major version design changes.

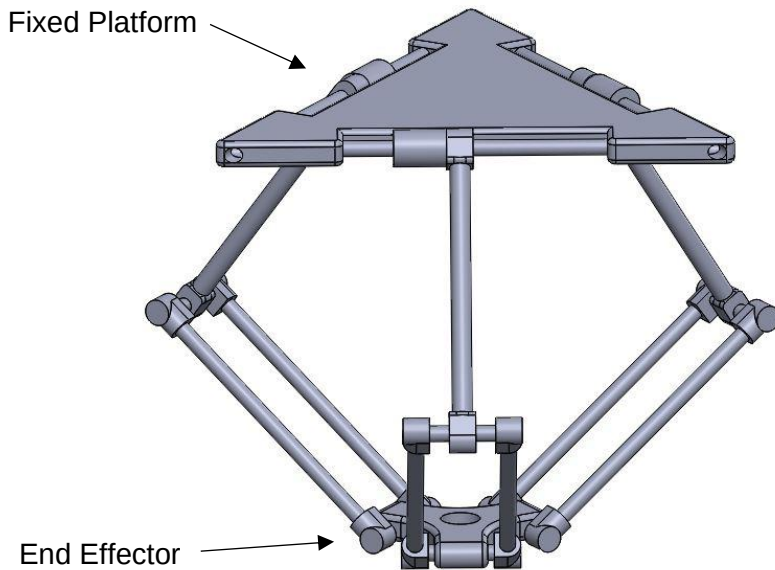


Figure 6 - V.1 Model

Since the project specification requested that the robot be as self-contained and portable as possible, this robot needed to be approached from a different angle, hence the orientation was flipped so that the moving platform is held above the control panel.

In converting the robot into a self-contained device, the base was split into three parts so the electronic components can be concealed within the base. Figure 3 shows the model of the Version 2 design, which is the first version to implement the upside-down design change and the compartmentalised base.

A Delta robot typically consists of the moving platform suspended down underneath the robot's control panel. Figure 6 shows the first model created in the project, a design heavily inspired by a typical pick-and-place delta robot. This version 1 design was designed with the intention of being held up by a tripod, attached at the top panel, and would have not been self-contained or easily portable.

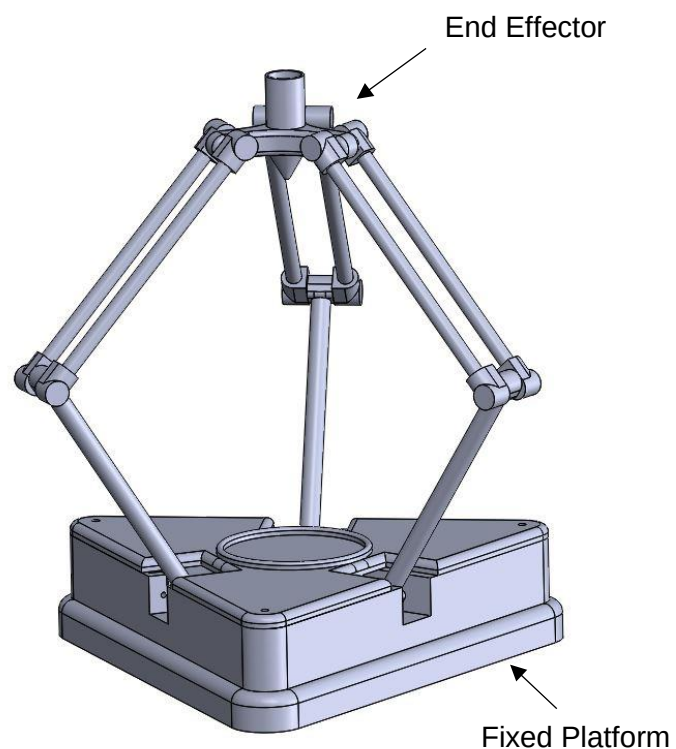


Figure 7 - V.2 Model

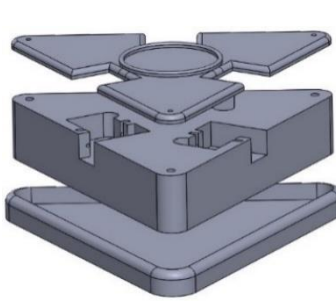


Figure 8 – Exploded View of V.2's Base Pieces

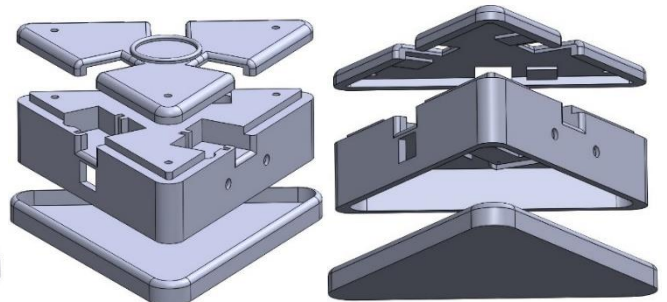


Figure 9 – Exploded View of V.5's Base Pieces

The compartmentalised base, consisting of three parts (base, base lid and base tray), was one of the components which went through the largest number of significant alterations during the prototyping phase. Figure 8 shows an exploded view of Version 2's base piece design, while Figure 9 shows the base pieces for the final version, V.5.

There are seven major changes between the two iterations of the base parts shown in figures 8 and 9. These are as follows:

1. The rim slot in V.5 allows the lid and base parts to fit together without the need for pins to hold the lid onto the base, unlike in V.2 where there are holes in the corners of the lid and base pieces for pins to slot in.
2. V.2 had no gaps in the walls of the Pi mount on the underside of the base piece to allow space for output cables from the raspberry pi. V.5 has gaps for all the output ports around the wall of the mount.
3. In V.2, the Pi Mount also did not allow space for the GPIO pins, which is why the Pi Mount in V.5 has a recessed panel to allow the right-angled pins to fit inside.
4. Recesses were added to the sides of the servo holes, to allow the servo cables to reach their holes to the raspberry pi.
5. In V.2, the servos were screwed into the walls of the servo slots. However, this would have caused assembly to be fiddly and the close proximity of the screw holes to the edge of the surface meant that prints tended to fail and leave a gap in the wall. In V.5, the servo's wings slot into gaps in the base piece to hold them in position instead.
6. In V.5, panels are extruded from the underside of the base lid to push down onto the servos to hold them in place.
7. Holes were added to the walls of the base for the power cable to go through and the button to screw onto.

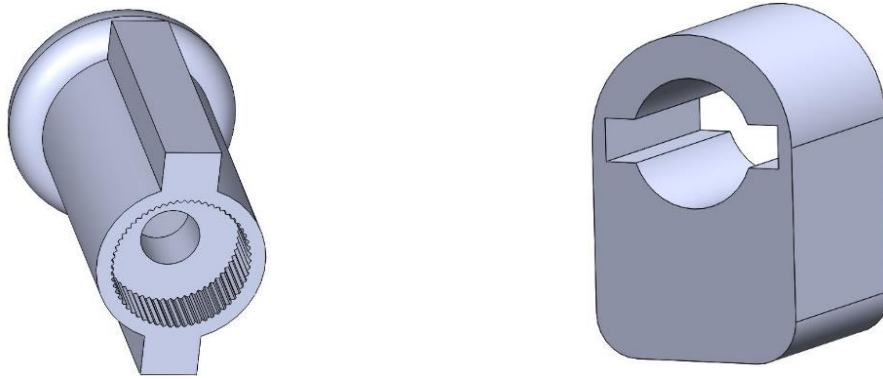


Figure 10 – V.3's Motor Pin and Motor Corner

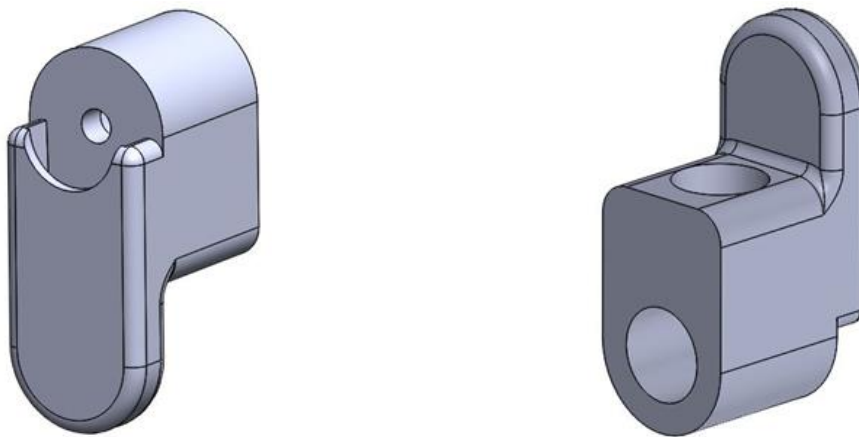


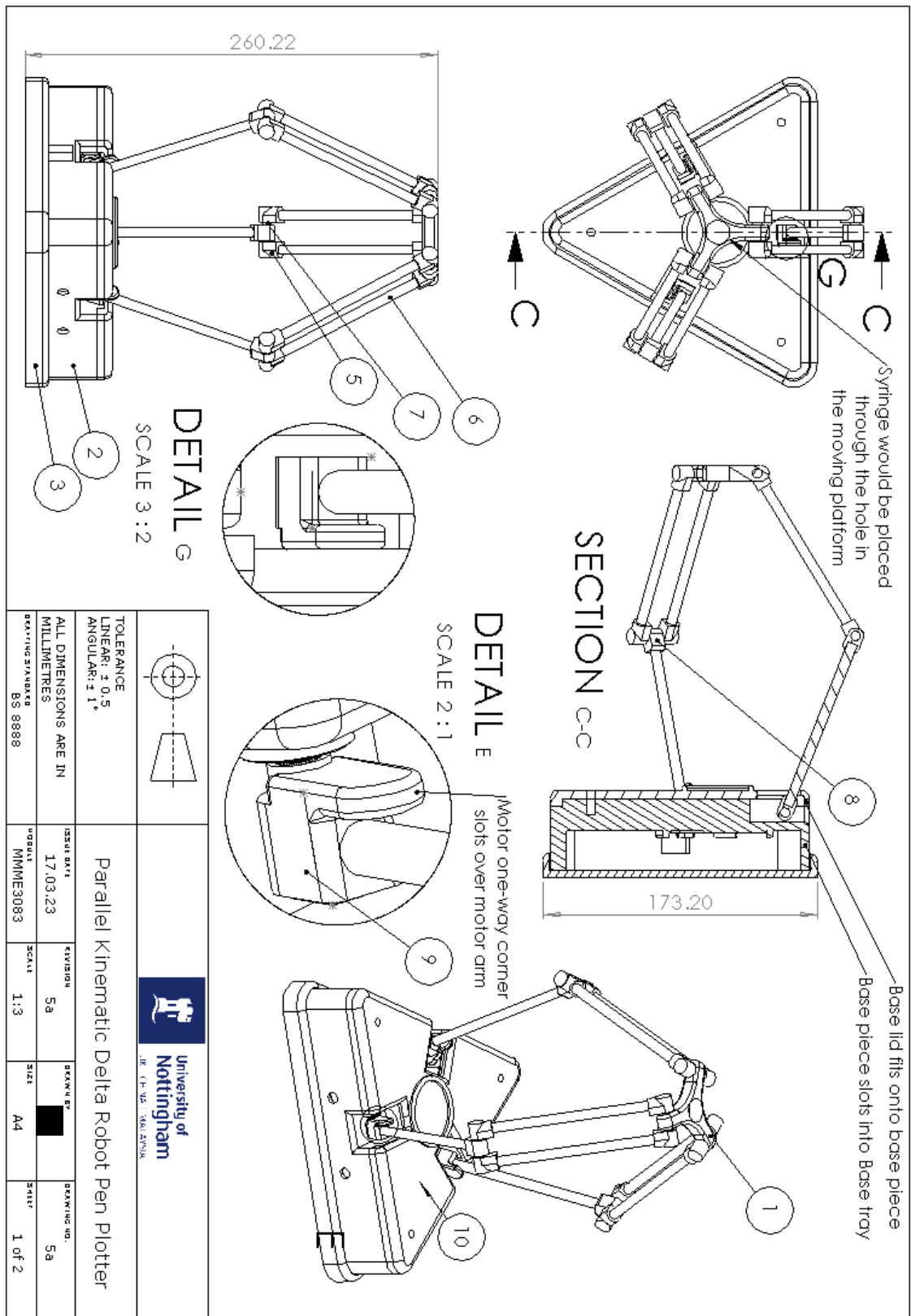
Figure 11 - V.5's Motor Corner

Another significant design challenge was how to attach the moving arms to the servo motor's moving tip to allow the servos to control the movement of the arms. Initially, in V.2 and V.3, this job was going to be done by the pieces shown in Figure 10. The motor pin would slot through the motor corner piece, kept in place torsionally by the aligning hole shape, and fit onto the motor head with matching splines. However, upon manufacturing the first test print for this piece, it became clear that the Nylon 12 powder did not have the necessary resolution to print the splines, so the motor pin would not attach to the motor head. Therefore, a redesign was necessary.

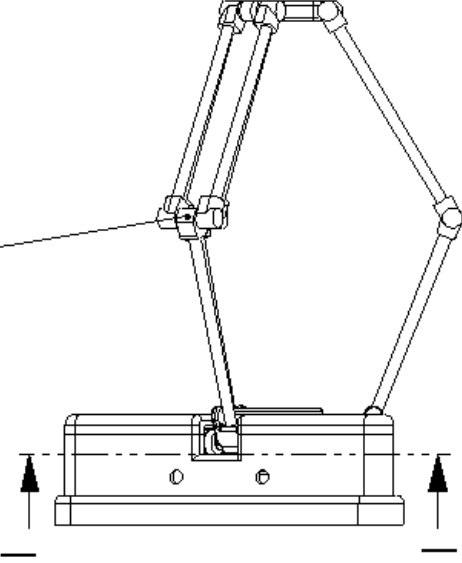
Deciding to make use of the motor head arm (Figure 11) that came with the servo motors and fitted perfectly onto the motor head, V.5's motor corner piece uses a sleeve that fits over the top and can also screw directly onto the motor head itself, so it is kept securely in place both torsionally and axially. Figure 11 shows the motor corner piece from the final version, V.5.

The General Assembly drawing of the robot is pictured on the following page.

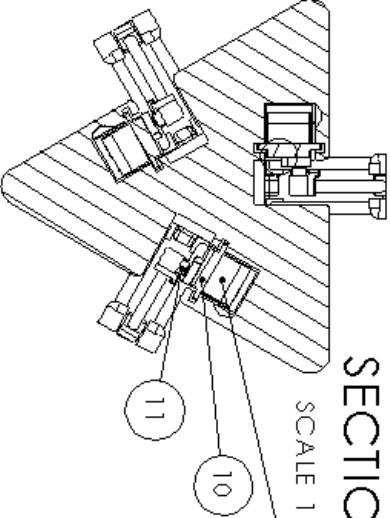
Please find the detail drawings for all the robot's parts in Appendix 1.



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Moving Platform\Assembly 5a	Moving Platform	1
2	Base Part\Assembly 5a	Base	1
3	Base Tray\Assembly 5a	Base Tray	1
4	Base Lid\Assembly 5a	Base Lid	1
5	Two-way corner\Assembly 5a	Two-way corner	12
6	long rod\Assembly 5a	Long rod	9
7	short rod\Assembly 5a	Short rod	6
8	one-way corner\Assembly 5a	One-way corner	3
9	motor one-way corner\Assembly 5a	Motor One-way corner	3
10	Servo Motor\Assembly 5a	Servo Motor	3
11	Motor Head arm\Assembly 5a	Motor Head arm	3



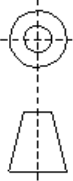
One-way corner piece slides onto the short rod between the two two-way corner pieces



SECTION H-H

SCALE 1 : 3

Motor fits into motor slots in the base piece

	TOLERANCE LINEAR: ± 0.5 ANGULAR: $\pm 1^\circ$			
ALL DIMENSIONS ARE IN MILLIMETRES				
DESIGN STANDARDS BS 8888	ISSUE DATE 17.03.23	REVISION 5a	DESIGNED BY 	DRAWING NO. 5a
PROJECT NAME MMME3083		SCALE 1:5	SHEET A4	SHEET 2 of 2



University of Nottingham
SCHOOL OF ENGINEERING

Parallel Kinematic Delta Robot Pen Plotter

3.2. Materials and Manufacturing

During the manufacturing phase of the project, the part with the most debate over the material to be used was the rods to create the legs. Five different rod materials were tested for the legs. These were 6mm diameter solid silver steel, 8mm diameter hollow silver steel, Carbon fibre and Composite rods. Wooden dowel and 3d printed PLA filament were also considered, but were eliminated before testing could take place because they were determined less likely to be successful due to the possibility of failure resulting from lack of stiffness. Eventually, carbon fibre was chosen as the most appropriate material due to its impressive stiffness despite being very light. It has a modulus of elasticity of 234 GPa and density of 1.55 g/cm³. The rods were then machined to the right lengths and tapered on a lathe.

The electronic components were all purchased separately, and all remaining parts were designed to be 3D printed. The corner parts to connect the rods on the arms all needed very precise dimensions, so the Nylon 12 powder printer was chosen to fabricate these parts. The three base pieces were selected to be printed in PLA since precise dimensions were less important for the system to work, and their larger geometries meant that using the more expensive Nylon 12 material would cause the project to exceed the budget of manufactured parts of £60. Since all electronic components were provided, this budget only has to cover the costs of the manufactured parts. The four 200mm length 6mm diameter carbon fibre rods cost a total of £15.98, leaving a remaining budget of £44.02 for the 3D printed parts.

Nylon 12 powder has an average price of £100 per kg and a density of 1.15g/cm³. PLA has an average price of £15 per kg and a density of 1.24g/cm³. The material cost calculations are shown below. The part names are used in accordance with the names they are given in the General Assembly Drawing above.

Corner Parts:

Motor One Way Corner: Volume 1300mm³(× 3)

Two way corner: Volume 1032mm³(× 12)

One way corner: Volume 951mm³(× 3)

Moving Platform: Volume 7019mm³(× 1)

Nylon 12 Density = 1.15g per cm³, price £100 per kg

$$(1300 \times 3) + (1032 \times 12) + (951 \times 3) + (7019 \times 1) = 26156\text{mm}^3$$

$$1.15\text{g per cm}^3 = 1.15 \times 10^{-3}\text{g per cm}^3$$

$$(1.15 \times 10^{-3}) \times 26156 = 30.0794\text{g}$$

£100 per kg means £0.1 per g

30g at £0.1 per g is £3

Base Parts:

Base: Volume $3.26 \times 10^5 \text{mm}^3$

Base Tray: Volume $1.21 \times 10^5 \text{mm}^3$

Base Part: Volume $9.23 \times 10^4 \text{mm}^3$

$$(3.26 \times 10^5) + (1.21 \times 10^5) + (9.23 \times 10^4) = 539300 \text{mm}^3$$

$$(1.15 \times 10^{-3} \text{g per mm}^3) \times 539300 = 620.195 \text{g}$$

620g Nylon 12 at £0.1 per g = £62 (Over budget)

PLA has density $1.24 \times 10^3 \text{g per mm}^3$

$$539300 \times 0.00124 = 668.732 \text{g}$$

668g PLA at £0.015 per g = $668 \times 0.015 = £10.02$

This gives a total cost of 3D printed parts of £13.02, which is well within the budget.

The Table of materials below lists each of the parts used in the assembly and which material was used for each of the manufactured parts.

Table 1 – Table of Materials

NYLON 12 (powder printer)	PLA (filament printer)	Carbon Fibre (Machined)	Electronic Components (purchased)
• 3x One-way corner • 12x Two-way corner • 3x Motor one-way corner • 1x Moving Platform	• 1x Base • 1x Base Lid • 1x Base Tray	• 6x 22mm rod • 9x 110mm rod	• 1x Raspberry Pi Zero • 3x FITEC Micro Servo 9g FS90MG • 1x PCB board (tbc) • 1x Ribbon Cable • 1x DC to mains outlet plug • 1x DC to open ended power cable • 2x jumper wires • 1x breakout board

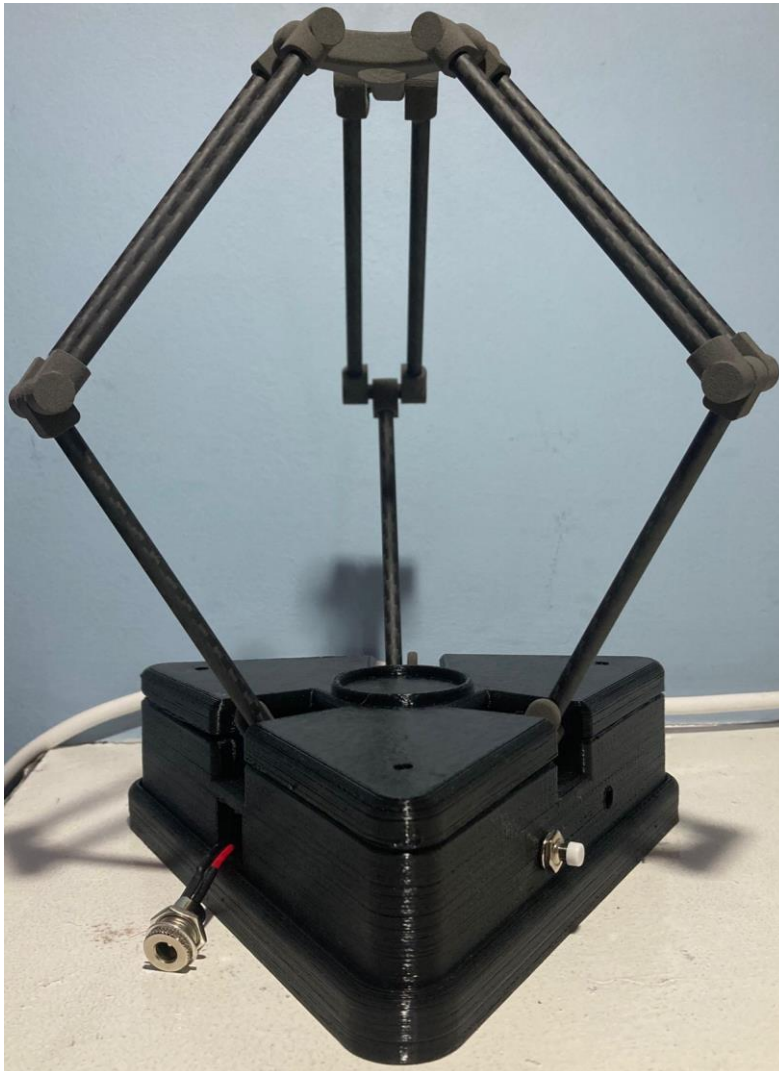


Figure 12 – Manufactured Robot based on V.5 model

Figure 12 shows the completed manufactured robot based on the 5th design iteration.

3.3. Electronics

In order to approach the electronic component design, it is necessary to know what in the assembly is going to be controlled electronically. Each of the three arms in the delta robot will be controlled by one servo motor. The assigned motor for use on this project was the Tower Pro Micro Servo 9g. The arms must each be attached to the turning pin of the motor, which is controlled to turn the arms to their necessary positions. Therefore, the 3 motors need to be controlled by something.

There were 3 different microcontrollers considered to control the actuators. These were the Raspberry Pi Pico, Raspberry Pi Zero and Raspberry Pi 4. While the original system design had the Raspberry Pi Pico as its intended microcontroller, upon further research, the Pico proved to be lacking in processing power and memory. The Pi 4 and Pi Zero both had more appropriate power and memory, with the 4 also having a built-in USB port, eliminating the need for an external 'on-the-go' cable.

For the final version, the Zero was chosen due to the similar form factor to the Pi Pico while simultaneously having a more appropriate processing power and memory.

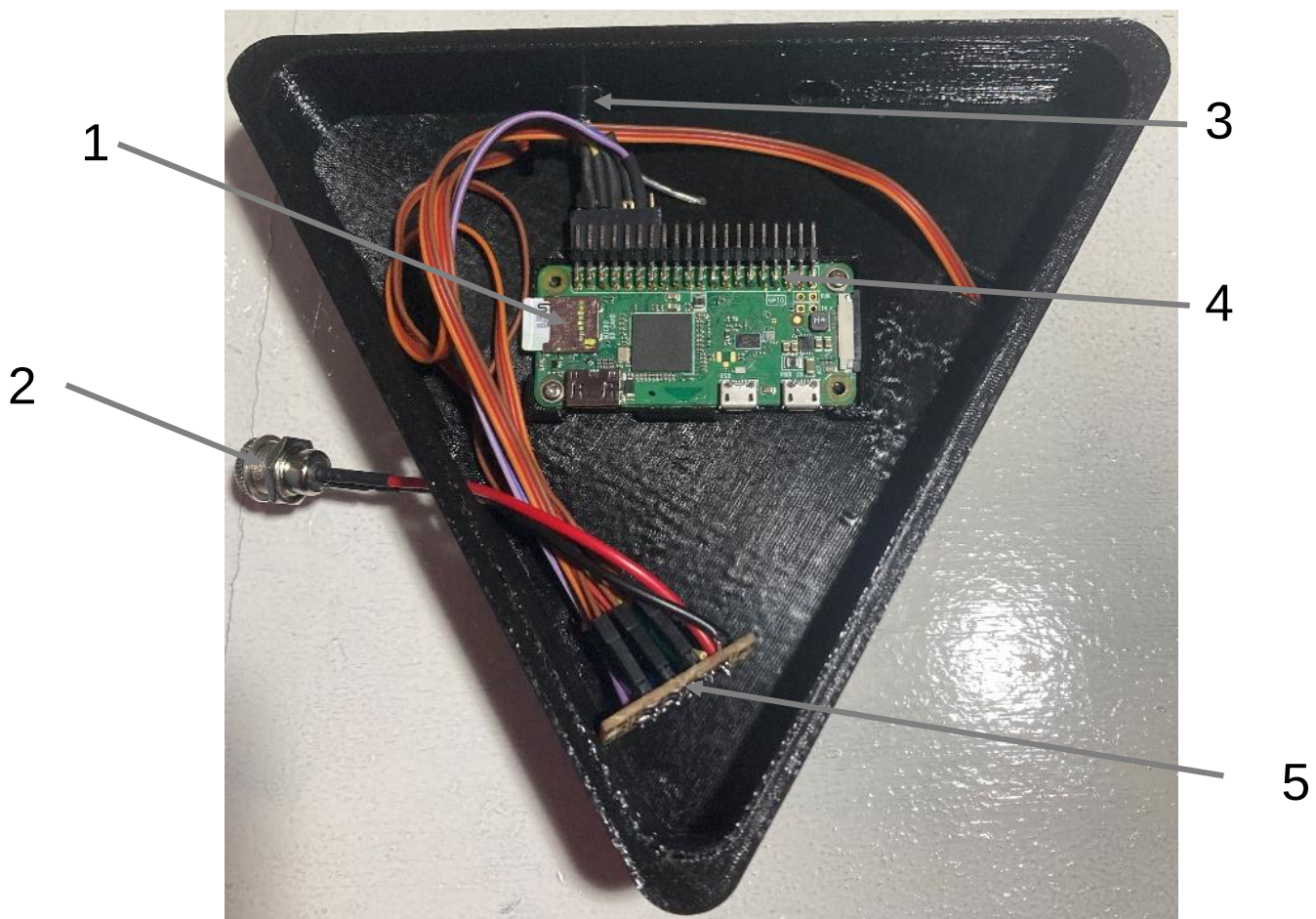


Figure 13 – Electrical Components in Base

Figure 13 shows the base part exposing the circuitry inside. The electronic system is comprised of the following components:

1. The Raspberry Pi Pico is the control board of the system. It is the green rectangular board in the centre of the base.
2. The power cable connects to a DC power cable which plugs into a mains outlet to provide the robot with power. It is the thick cable with a black and a red wire.
3. The Start/stop button connects to the raspberry Pi and is programmed so that when the button is pushed, the servo motor control program will start to run. Pushing it again will cause the program to abort. Although the head of the button is not visible in Figure 13, the inside mount of the button and its cable are visibly screwed into the base wall midway across the wall closest to the top of Figure 13.
4. The GPIO pins along the top of the Raspberry Pi Pico are used to connect the wires in the circuit to connect each component together. They are right-angle GPIO pins so that the wires can extend from the side of the Raspberry Pi rather than underneath, where they could collide with the base pieces.
5. The breakout board is necessary to connect the servo motors to the main control board and to a power source. It connects everything in the circuit together, attaches to the Pi zero, each servo motor, and the power cable. It is the small board in the bottom corner of the base piece in the picture.

3.4. Reverse and Forward Kinematic Model

In order to determine the necessary movements of the servo motors in the raspberry pi control system, we must know the desired positions of the arms and the end effector and the corresponding angles of each joint. Kinematic analysis of components in delta robots has an extensive covering of research. 'Delta robot: inverse, direct, and intermediate Jacobians' [8] explores the kinematic analysis of parallel manipulators using Jacobian matrices. This approach helps to identify singular configurations. Another approach is demonstrated in 'An application of screw theory to the kinematic analysis of a Delta-type robot' [9], which outlines how to find a "closed-form solution by applying a new strategy based on the unknown coordinates of a point embedded to the moving platform". For this project, however, the kinematic calculations follow the standard method of kinematic analysis demonstrated in the Trossen Robotics Community's 'Delta Robot Kinematics' [6] tutorial.

Figure 14 shows a kinematic scheme for a basic 3 DOF Delta robot. Since it models a standard delta robot, which has an end effector suspended below the fixed base rather than the base being fixed below the end effector, this model is upside-down compared to the robot in this project. However, the mathematical analysis remains the same. The two platforms are equilateral triangles. The green triangle represents the fixed platform with the motors, and the pink triangle represents the moving platform with the end effector. The Joint angles are θ_1 , θ_2 and θ_3 , and point E0 is the end effector position with coordinates (x_0, y_0, z_0) . In order to solve the inverse kinematics problem, a function with E0 coordinates (x_0, y_0, z_0) as parameters which returns $(\theta_1, \theta_2, \theta_3)$ must be created. The forward kinematics functions get $(\theta_1, \theta_2, \theta_3)$ and return (x_0, y_0, z_0) .

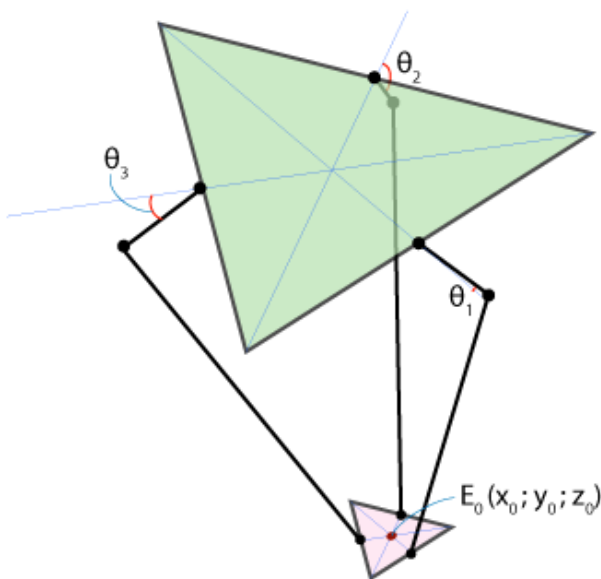


Figure 15 - Basic Delta Robot Kinematic scheme from Trossen Robotics Community [6]

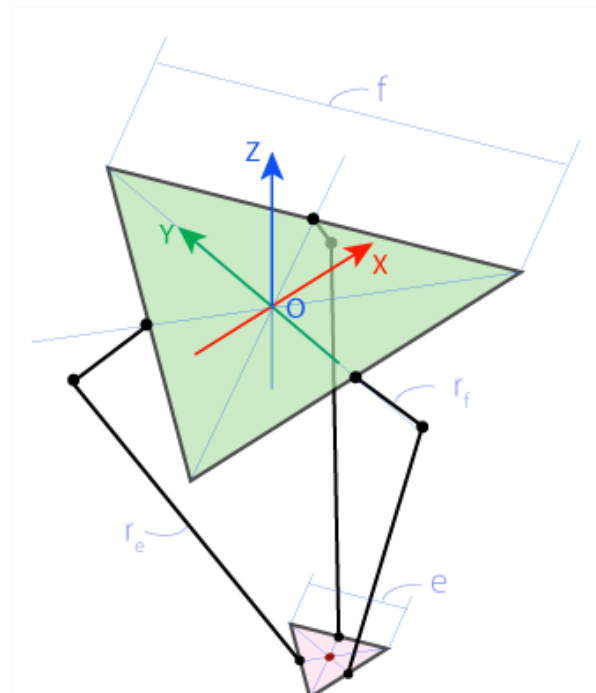


Figure 14 - Basic Delta Robot Kinematic Model Reverse Kinematic Parameter Definition Diagram from Trossen Robotics Community [6]

Figure 15 shows the variables assigned to the system's parameters in the robot geometry. The side of the fixed platform is f , the side of the end effector platform is e , the length of the upper arm is rf and the length of the lower arm is re . For the Delta robot in this project, these values are as follows:

$$\begin{aligned} f &= 147mm \\ e &= 85.98mm \\ rf &= 120mm \\ re &= 120mm \end{aligned}$$

The reference frame is at the origin of the centre of symmetry of the fixed triangle, so the end effector's z-coordinate is negative. The geometry represented by f , e , rf and re in this project's delta robot is shown in Figures 16 and 17. The construction triangle in Figure 16 is of side length e , and the three lines in the triangle are each colinear with the axes in the centre of the holes through which the rectangular arm goes. The construction triangle in Figure 17 is of side length f , and the three lines in the triangle are each colinear with the axes in the centre of the holes through which the motor corner piece screws onto the motor head. The two blue construction lines in figure 18 represent re and rf , and are of equal length.

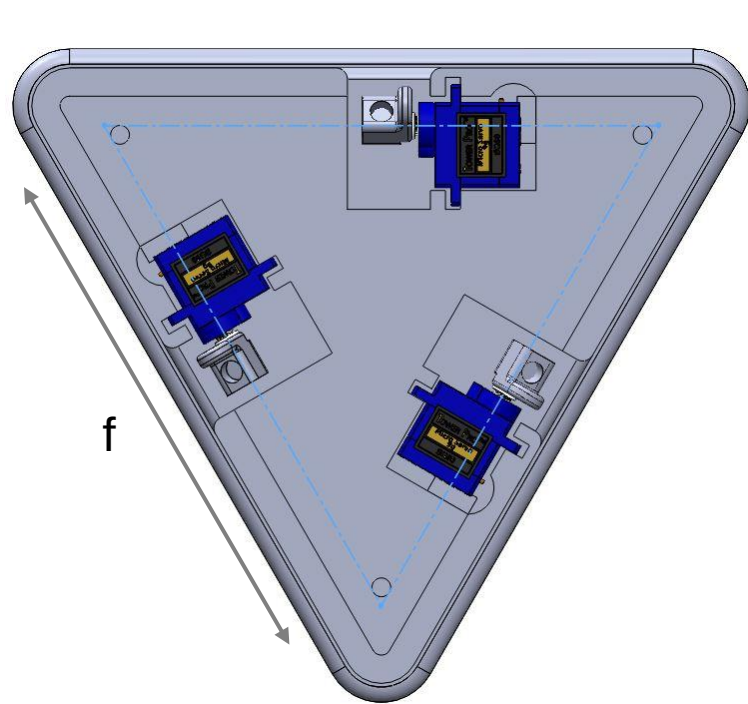


Figure 16 - pictural representation of value f in the system, blue construction equilateral triangle has side length = $f = 147mm$

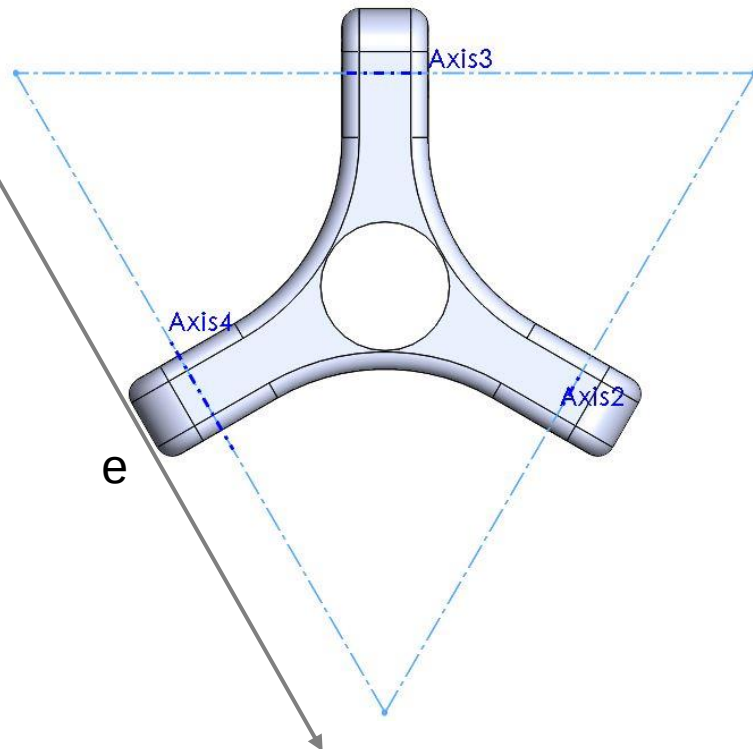


Figure 17 - pictural representation of value e in the system, blue construction equilateral triangle has side length = $e = 85.98mm$

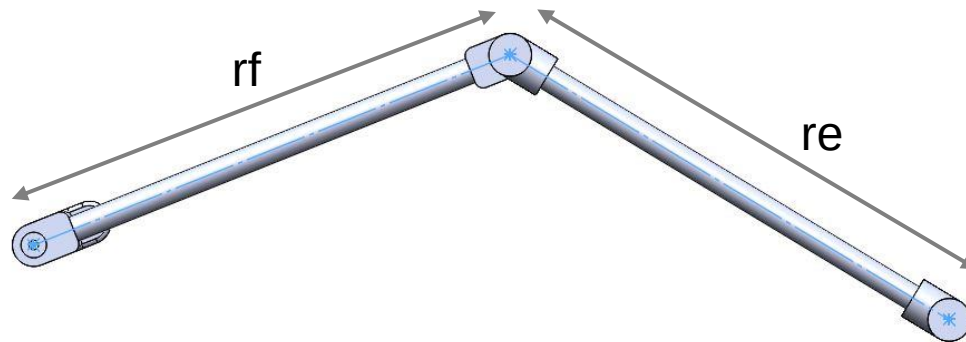


Figure 18 - pictural representation of values rf and re in the system, blue construction lines are length = $rf = re = 120\text{mm}$

Reverse Kinematics:

As pictured in Figure 19, Joint F1J1 can only rotate in the YZ plane, forming a circle with centre point F1 and radius rf . As opposed to F1, J1 and E1 are 'universal joints', which means that E1J1 can rotate freely relative to E1, forming a sphere with centre point E1 and radius re .

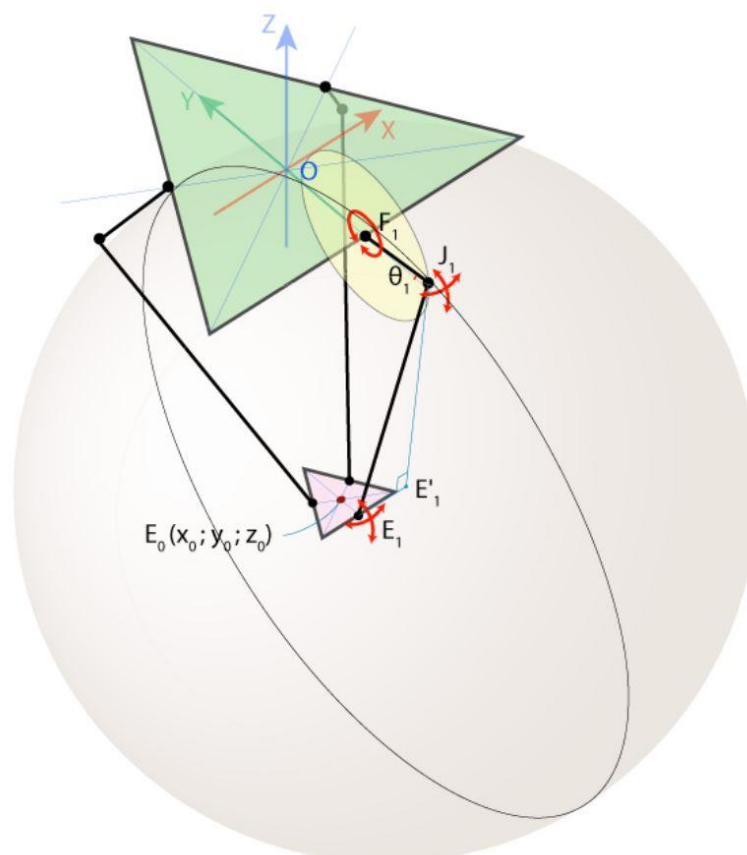


Figure 19 - Rotational Analysis Diagram from Trossen Robotics Community [6]

The intersection of this sphere and the YZ plane is a circle with centre point E'1 and radius E'1J1, where E'1 is the projection of the point E1 on the YZ plane. The point J1 becomes the intersection of two circles with centres in E'1 and F1. J1 can be used to calculate angle θ_1 :

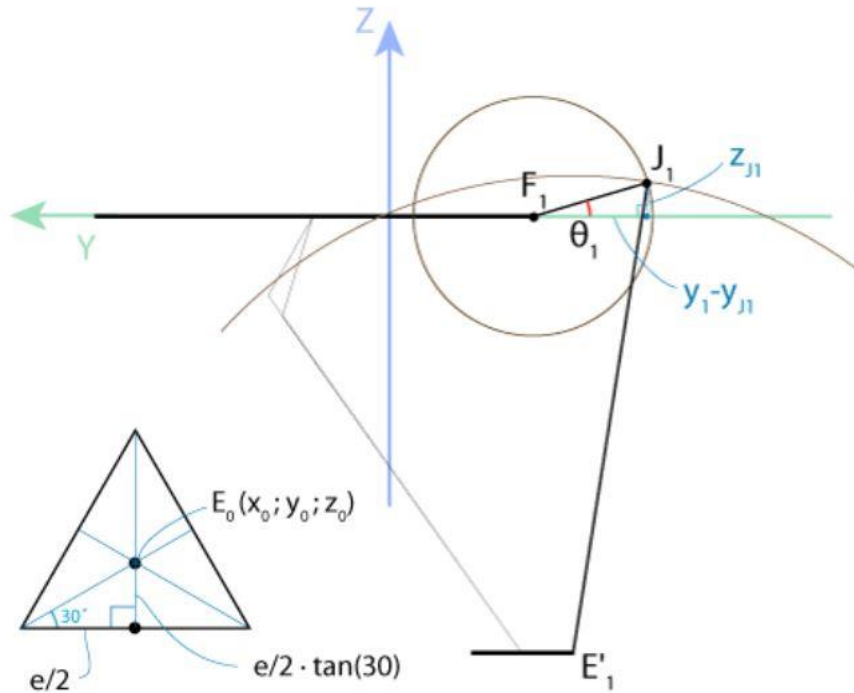


Figure 20 - YZ Plane View from Trossen Robotics Community [6]

$$\begin{aligned}
 E(x_0, y_0, z_0) \\
 EE_1 &= \frac{e}{2} \tan 30^\circ = \frac{e}{2\sqrt{3}} \\
 E_1 \left(x_0, y_0 - \frac{e}{2\sqrt{3}}, z_0 \right) &\Rightarrow E'_1 \left(0, y_0 - \frac{e}{2\sqrt{3}}, z_0 \right) \\
 E_1 E'_1 = x_0 &\Rightarrow E'_1 J_1 = \sqrt{E'_1 J_1^2 - E_1 E'_1^2} = \sqrt{r_e^2 - x_0^2} \\
 F_1 \left(0 - \frac{f}{2\sqrt{3}}, 0 \right)
 \end{aligned}$$

$$\begin{aligned}
 &\begin{cases} (y_{J1} - y_{F1})^2 + (z_{J1} - z_{F1})^2 = r_f^2 \\ (y_{J1} - y_{E1})^2 + (z_{J1} - z_{E1})^2 = r_e^2 - x_0^2 \end{cases} \\
 &\Rightarrow \begin{cases} (y_{J1} + \frac{f}{2\sqrt{3}})^2 + z_{J1}^2 = r_f^2 \\ (y_{J1} - y_0 + \frac{e}{2\sqrt{3}})^2 + (z_{J1} - z_0)^2 = r_e^2 - x_0^2 \end{cases} \Rightarrow J_1(0, y_{J1}, z_{J1}) \\
 &\theta_1 = \arctan \left(\frac{z_{J1}}{y_{F1} - y_{J1}} \right)
 \end{aligned}$$

Such algebraic simplicity follows from the choice of reference frame: joint F1J1 moves in the YZ plane only, so we can completely omit the X coordinate. The delta robot's symmetry can be used to determine the remaining angles, θ_2 and θ_3 .

First, the coordinate system must be rotated in the XY plane around the Z-axis through an angle of 120 degrees anti-clockwise, as is shown in Figure 21.

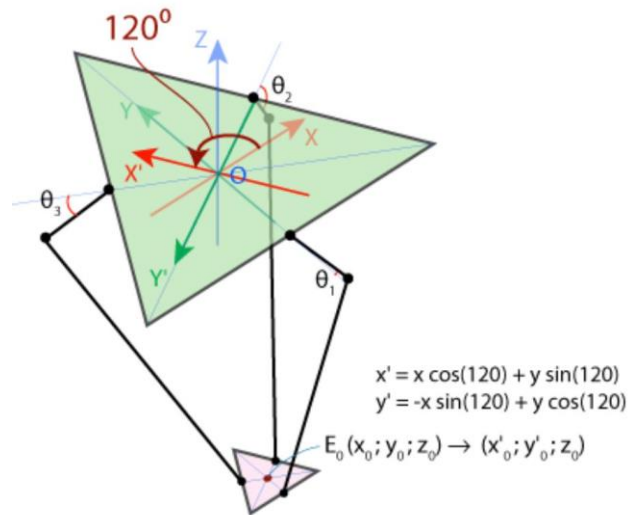


Figure 21 - Reference Frame Diagram from Trossen Robotics Community [6]

The new reference frame $X'Y'Z'$ can find angle θ_2 using the same algorithm used to find θ_1 . However, this time new coordinates x'_0 and y'_0 for the point E_0 must be determined using the corresponding rotation matrix. To find angle θ_3 the reference frame must rotate clockwise. The function calculates the angle θ for the YZ plane, and this function is repeated three times for each angle and each reference frame.

Forward Kinematics:

Having calculated angles θ_1 , θ_2 and θ_3 , the coordinates (x_0, y_0, z_0) of end effector point E_0 can be found. The angle values determine the points J_1 , J_2 and J_3 as seen in Figure 22. Joints J_1E_1 , J_2E_2 and J_3E_3 can freely rotate around points J_1 , J_2 and J_3 respectively, forming three spheres with radius re .

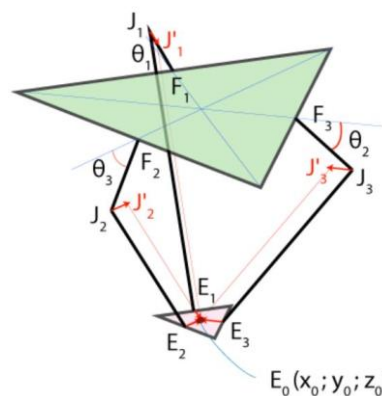
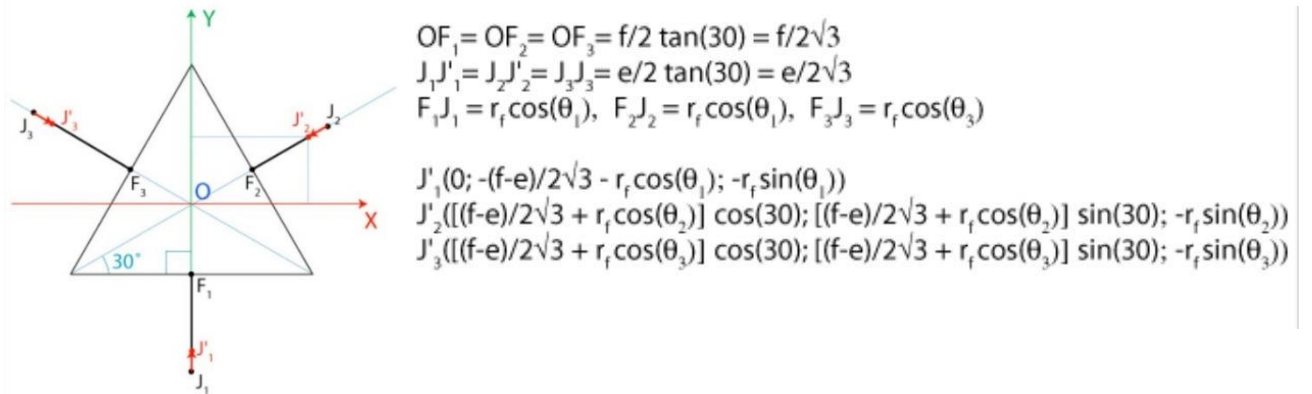


Figure 22 - Basic Delta Robot Kinematic Model Forward Kinematic Parameter Definition Diagram from Trossen Robotics Community [6]

Now the centres of the spheres should move from points J1, J2 and J3 to the points J'1, J'2 and J'3 using transition vectors E1E0, E2E0 and E3E0 respectively. After this transition all three spheres will intersect at point E0, shown in Figure 22.

To find the coordinates (x0, y0, z0) of point E0, three equations must be solved. To find coordinates of points J'1, J'2, J'3:

(Coordinates are points J1, J2, J3 as (x1, y1, z1), (x2, y2, z2) and (x3, y3, z3), x0=0)



$$\begin{cases} x^2 + (y - y_1)^2 + (z - z_1)^2 = r_e^2 \\ (x - x_2)^2 + (y - y_2)^2 + (z - z_2)^2 = r_e^2 \\ (x - x_3)^2 + (y - y_3)^2 + (z - z_3)^2 = r_e^2 \end{cases} \Rightarrow \begin{cases} x^2 + y^2 + z^2 - 2y_1y - 2z_1z = r_e^2 - y_1^2 - z_1^2 \\ x^2 + y^2 + z^2 - 2x_2x - 2y_2y - 2z_2z = r_e^2 - x_2^2 - y_2^2 - z_2^2 \\ x^2 + y^2 + z^2 - 2x_3x - 2y_3y - 2z_3z = r_e^2 - x_3^2 - y_3^2 - z_3^2 \end{cases} \quad (1)$$

$$w_i = x_i^2 + y_i^2 + z_i^2$$

$$\begin{cases} x_2x + (y_1 - y_2)y + (z_1 - z_2)z = (w_1 - w_2)/2 \\ x_3x + (y_1 - y_3)y + (z_1 - z_3)z = (w_1 - w_3)/2 \\ (x_2 - x_3)x + (y_2 - y_3)y + (z_2 - z_3)z = (w_2 - w_3)/2 \end{cases} \quad (4) = (1) - (2)$$

$$(5) = (1) - (3)$$

$$(6) = (2) - (3)$$

From (4)-(5):

$$x = a_1z + b_1 \quad (7)$$

$$y = a_2z + b_2 \quad (8)$$

$$a_1 = \frac{1}{d}[(z_2 - z_1)(y_3 - y_1) - (z_3 - z_1)(y_2 - y_1)] \quad a_2 = -\frac{1}{d}[(z_2 - z_1)x_3 - (z_3 - z_1)x_2]$$

$$b_1 = -\frac{1}{2d}[(w_2 - w_1)(y_3 - y_1) - (w_3 - w_1)(y_2 - y_1)] \quad b_2 = \frac{1}{2d}[(w_2 - w_1)x_3 - (w_3 - w_1)x_2]$$

$$d = (y_2 - y_1)x_3 - (y_3 - y_1)x_2$$

Figure 23 - Coordinate Diagram and Forward Kinematic Calculations
from Trossen Robotics Community [6]

(7) and (8) can be substituted into (1):

$$(a_1^2 + a_2^2 + 1)z^2 + 2(a_1 + a_2(b_2 - y_1) - z_1)z + (b_1^2 + (b_2 - y_1)^2 + z_1^2 - r_e^2) = 0$$

The final step is to solve this quadric equation and find z0, and then calculate x0 and y0 from eq. (7) and (8).

3.5. Experimentation

Two experiments were carried out to test the motor control capabilities. The first test programme instructed the robot to move the end effector up and down, while the second instructed the robot to move the end effector from side to side.

Figures 24 and 25 present time-stamped screenshots from the videos taken of the motor control experiments. These images show the paths followed by the end effector in the robot's first and second motor control tests.

In experiment 1, the programme commanded the servos to turn in such a way that the end effector was raised and lowered. The movement for this was less smooth than anticipated, due to the fact that the motor heads rotate in increments rather than one continuous rotational movement.

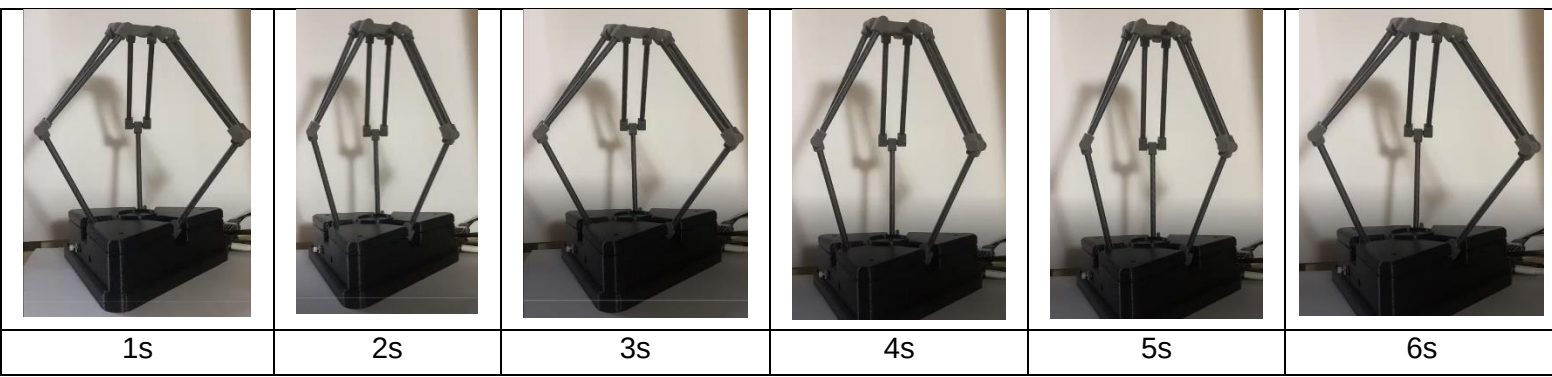


Figure 24 – Screenshots of Experiment 1 at one second increments

In experiment 2, the programme commanded the servos to turn in such a way that the end effector was tilted from side to side.

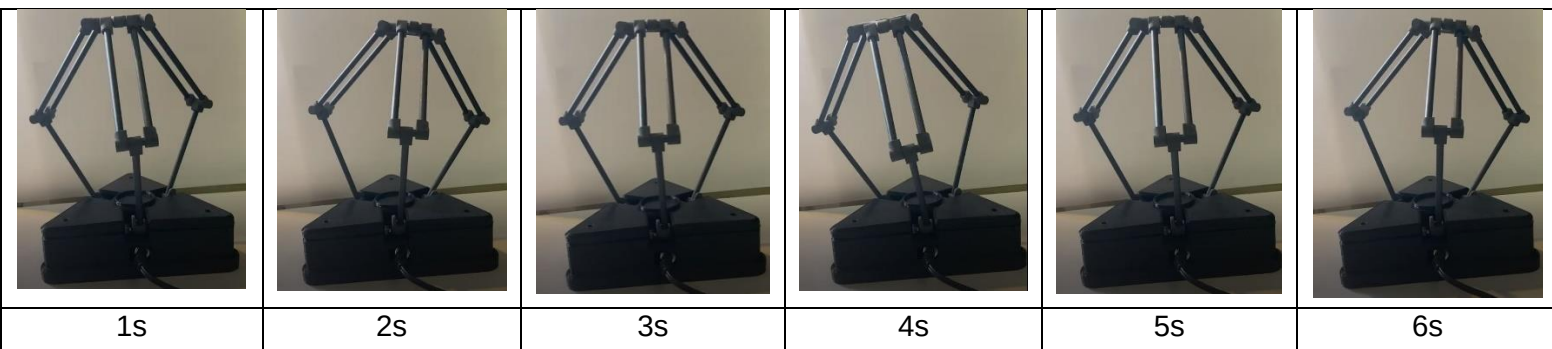


Figure 25 – Screenshots of Experiment 2 at one second increments

The code used to control the actuators in the two experiments can be found in Appendix 2.

Video recordings of these experiments are available at the following hyperlinks:

Experiment 1 - <https://www.youtube.com/shorts/aTT7BsTHM0M>

Experiment 2 - <https://www.youtube.com/shorts/ALtgyKC1spg>

3.6. Discussion

The system designed and manufactured in this project has some limitations compared to typical delta robots used in both professional and commercial contexts. One of the most impactful of these is the low precision of the actuators themselves. Because of the mechanical capacity of the pre-assigned motors for this project (FITEC Micro Servo 9g FS90MG), the motor heads cannot rotate smoothly and continuously. Instead, they rotate at set increments which causes the rotational movement to sometimes be jerky, which affects the shape formed by the path moved by the end effector. Without this limitation on the key electrical components, the patterns drawn out by the path of movement of the end effector could be significantly more complex.

This design, however, also comes with advantages. The small size of the pre-assigned motors means that the materials for the arms and end effector platform had to be very lightweight so that they could be moved by the limited power of the actuators. This meant that materials were relatively cheap to manufacture, compared to the steel rods that are usually used in place of the carbon fibre rods and 3D-printed corner pieces. Also, the self-contained design of the robot, with the end effector held above the fixed platform rather than suspended below it, means that the robot does not require a tripod to hold it in place. It is also easily portable and only needs to be plugged in with one cable at the wall.

4. CONCLUSIONS

To conclude, the project involved the design, manufacture, testing and application of a parallel kinematic delta robot to be used as an icing-ejecting pen plotter for an educational demonstration. The robot is compact and self-contained, as was the aim, and has been used in two motor control experiments. Despite proving the success of the robot's function, these experiments also raised a potential problem with the lack of precision in the servo motors leading to the path of the end effector being more difficult to control than anticipated. This could be improved by using actuators with capacity for higher precision.

At the conclusion of this project, a fully functional parallel kinematic mechanical system has been constructed, with the intention of using a Raspberry Pi programme which demonstrates the mechanical capabilities of the robot through instructing it to eject icing onto a cake in a pre-determined pattern. It has been the aim of the project to present an engaging robotic demonstration by appealing to common interests, in the hopes that, as Da Vinci suggested, this engagement of interest could spark inspiration in those who might otherwise have allowed the fascination of robotics to slip by without pursuing their understanding further.

5. POTENTIAL FUTURE WORK

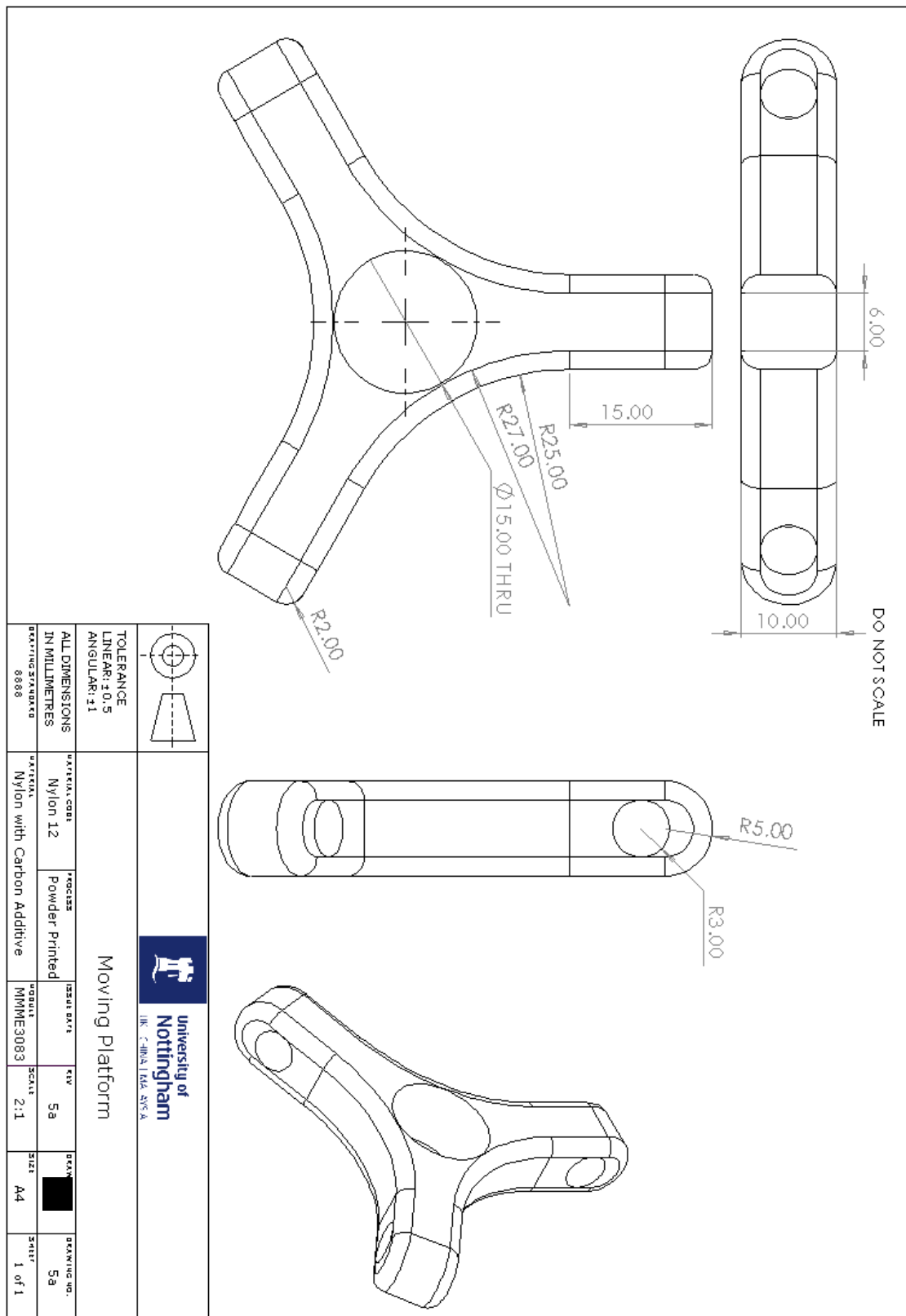
Potential further work for this project could include a more thorough servo control program, with a full decoration demonstration. Since the robot was designed to run a programme which would control the servos to perform a full decoration pattern for a cupcake, having included this in the project would have given more scope for the demonstration. However, this was not included in the project plan and criteria, and the time frame did not allow for optional extra work.

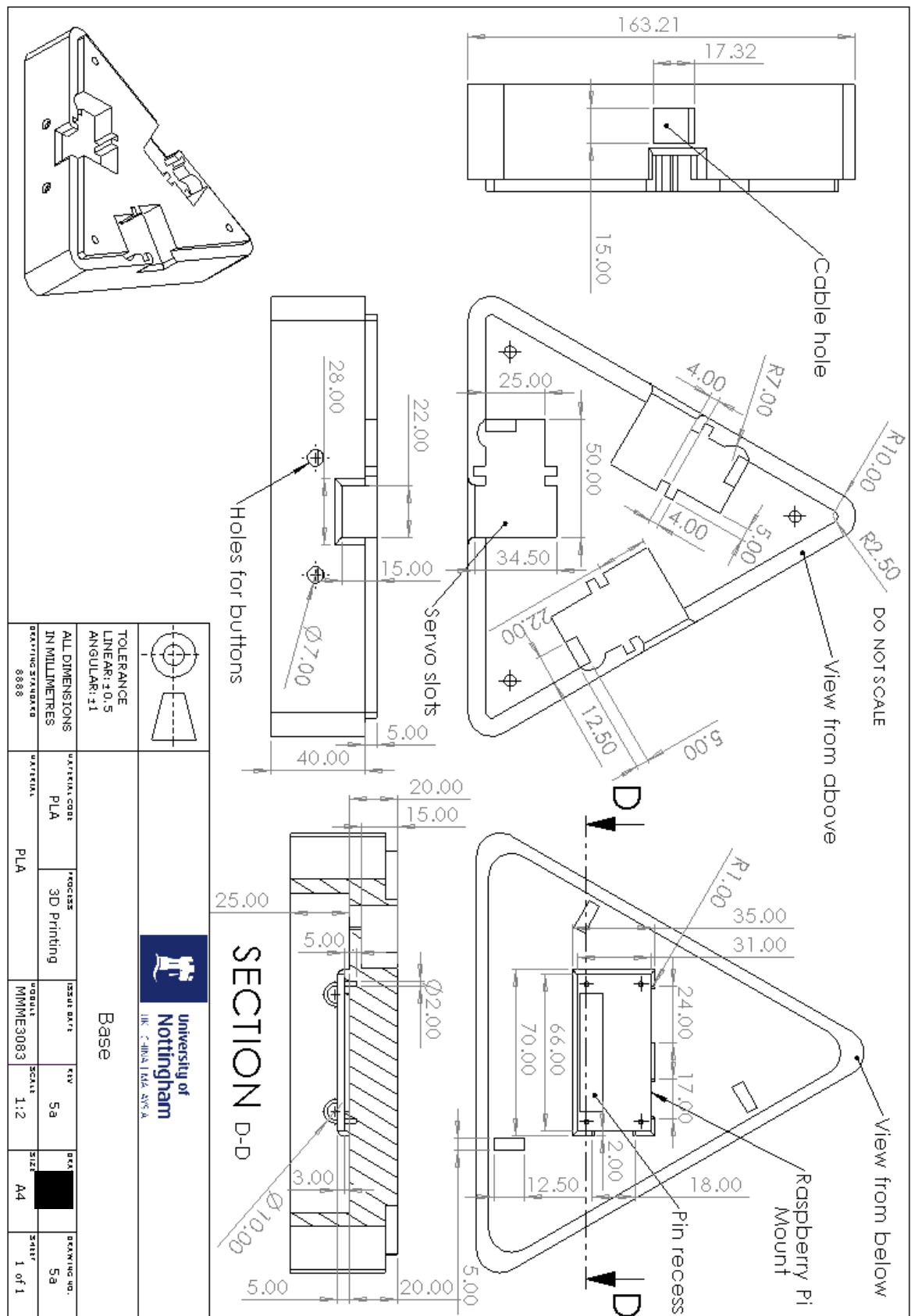
The icing ejection system could also be designed and added to the model, rather than just having an indication of how this would have worked on the main robot design. Had there been more time, these would have been part of the project itself.

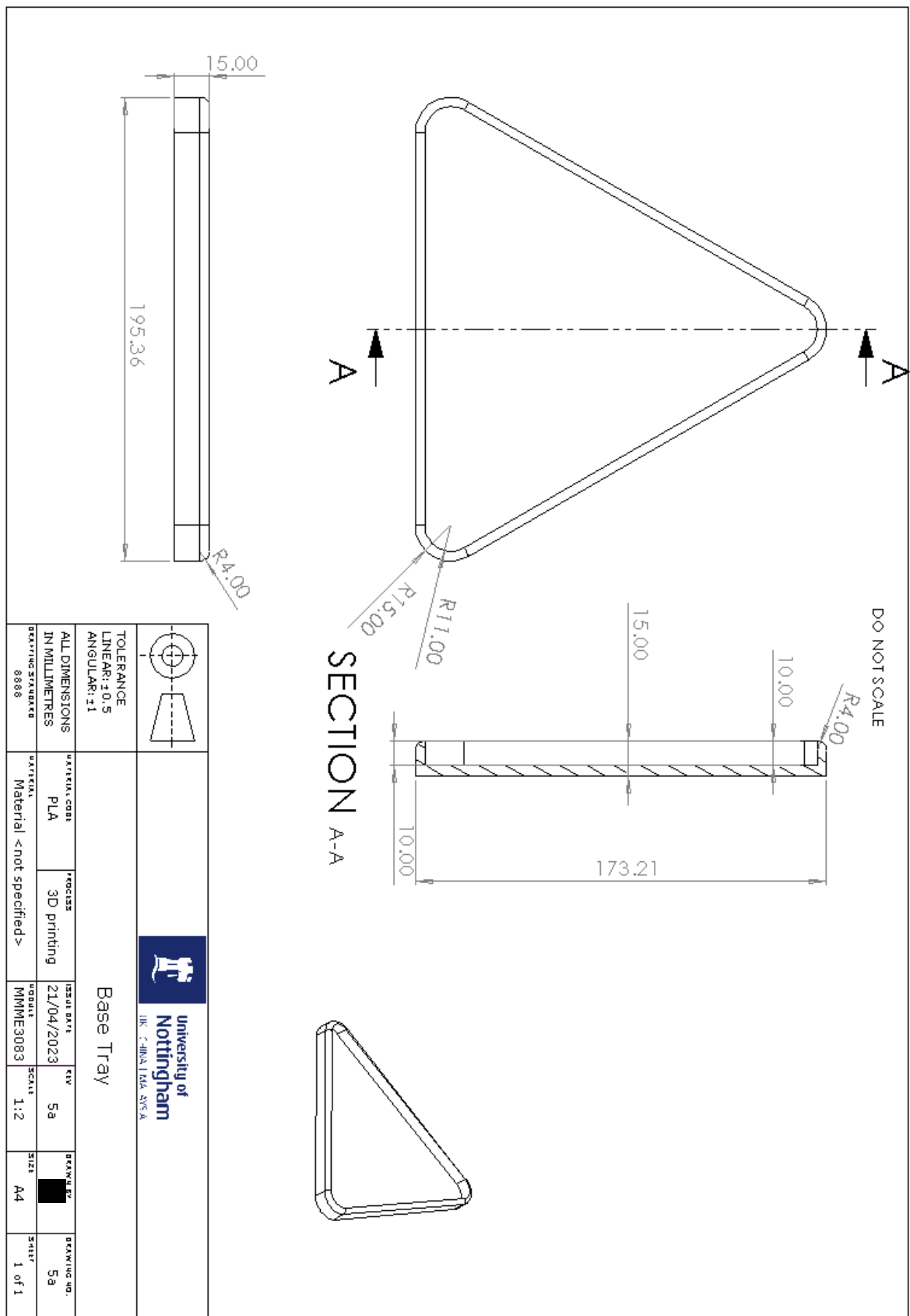
6. References

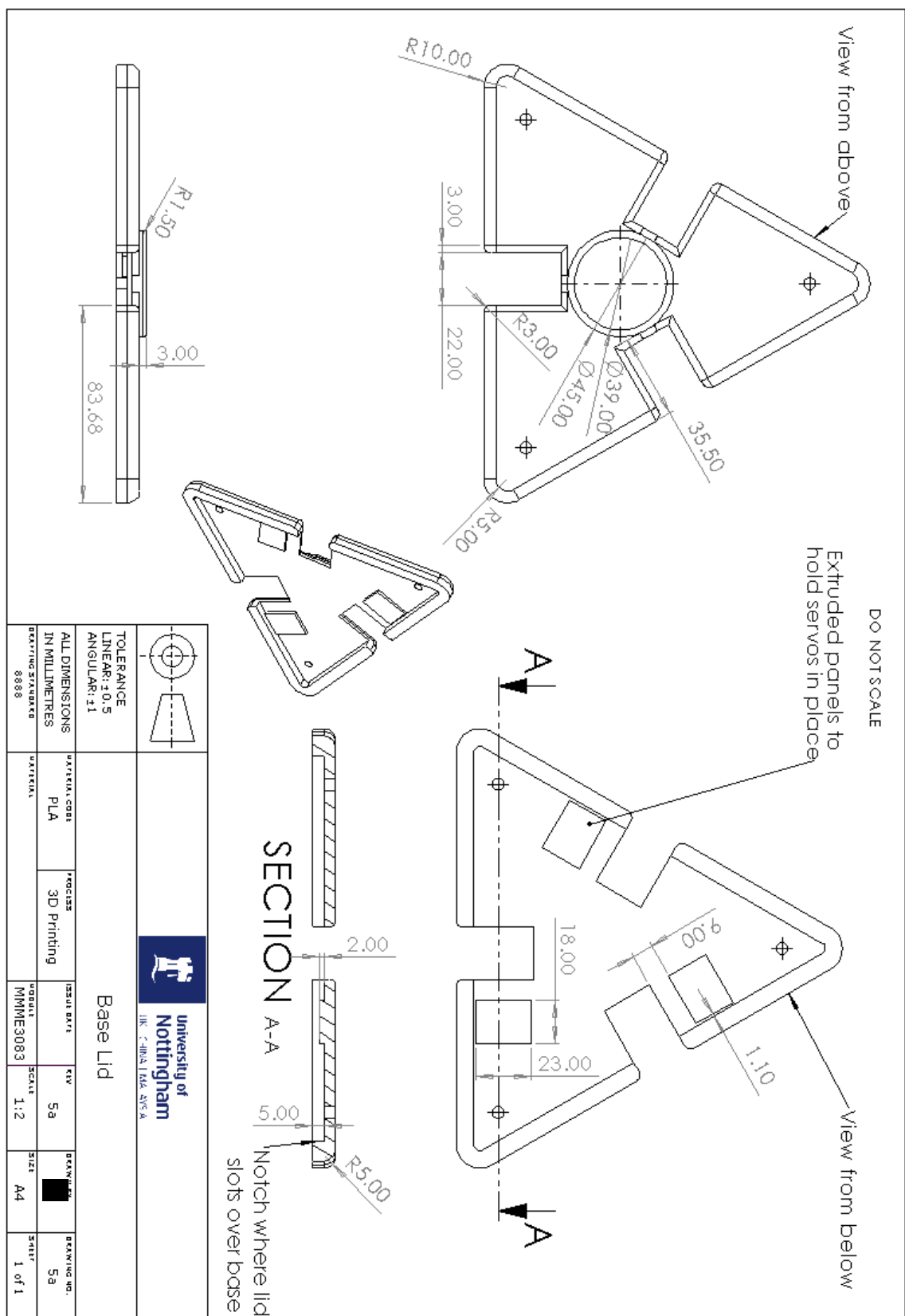
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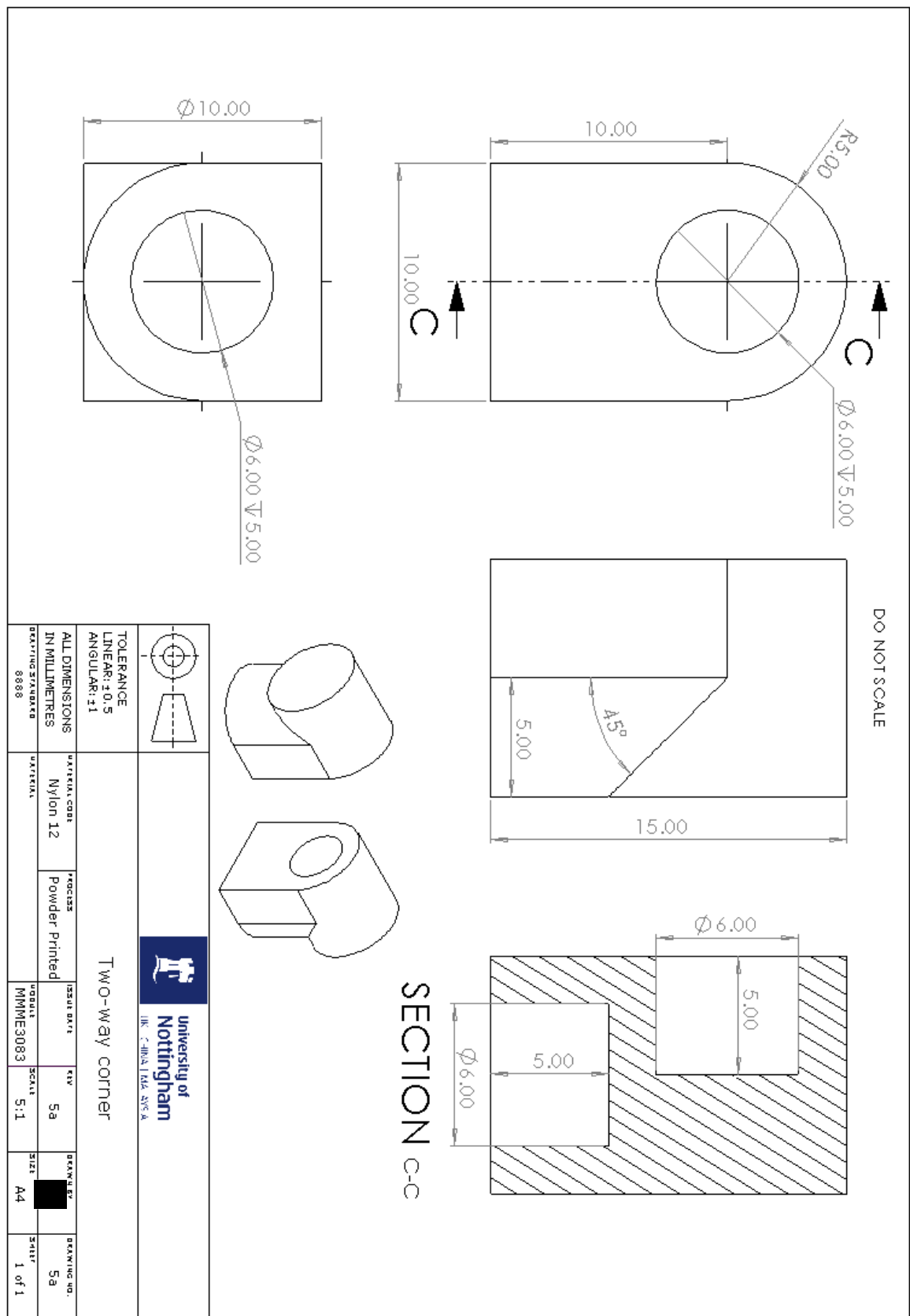
APPENDIX 1 – Robot Detail Drawings

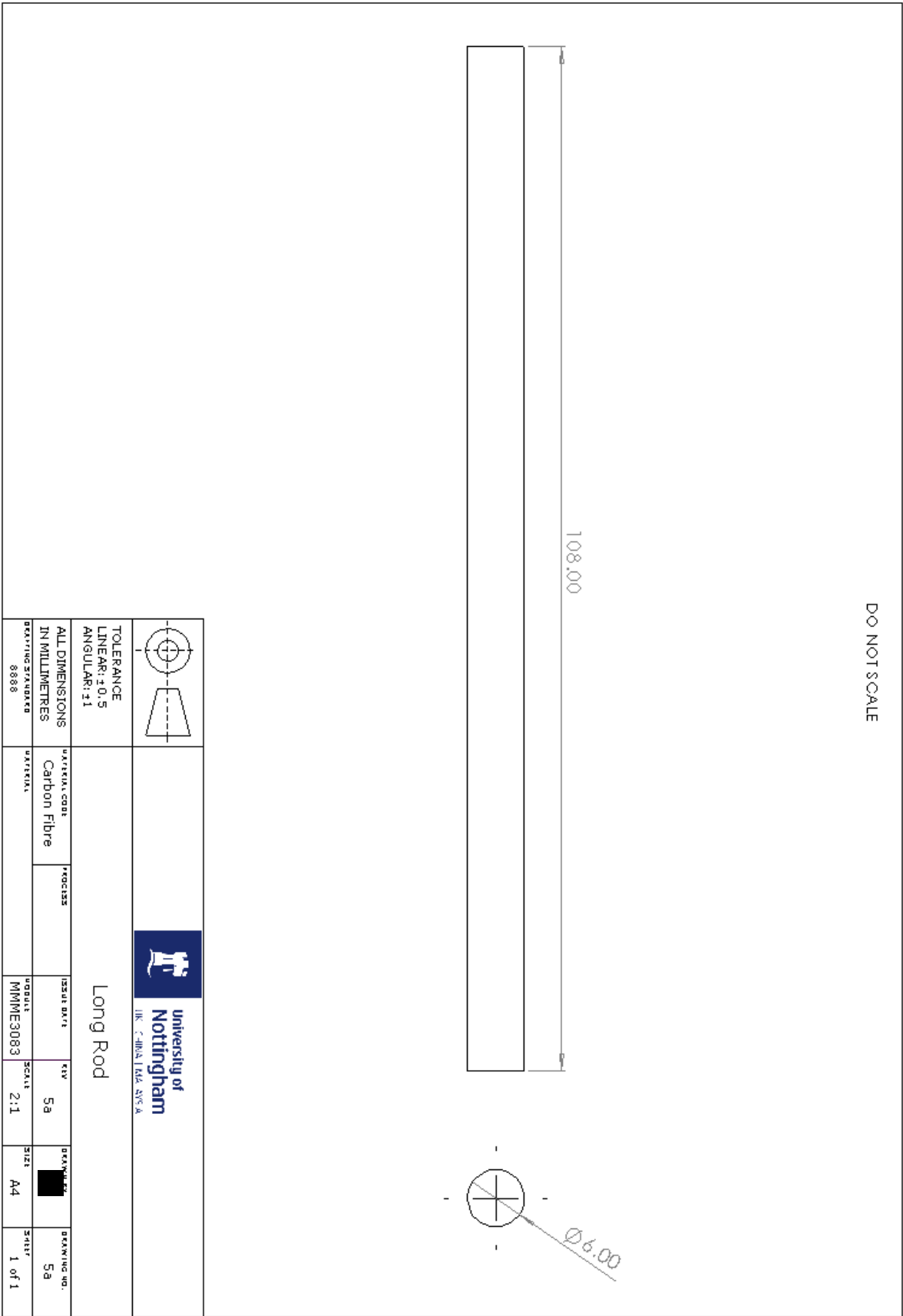


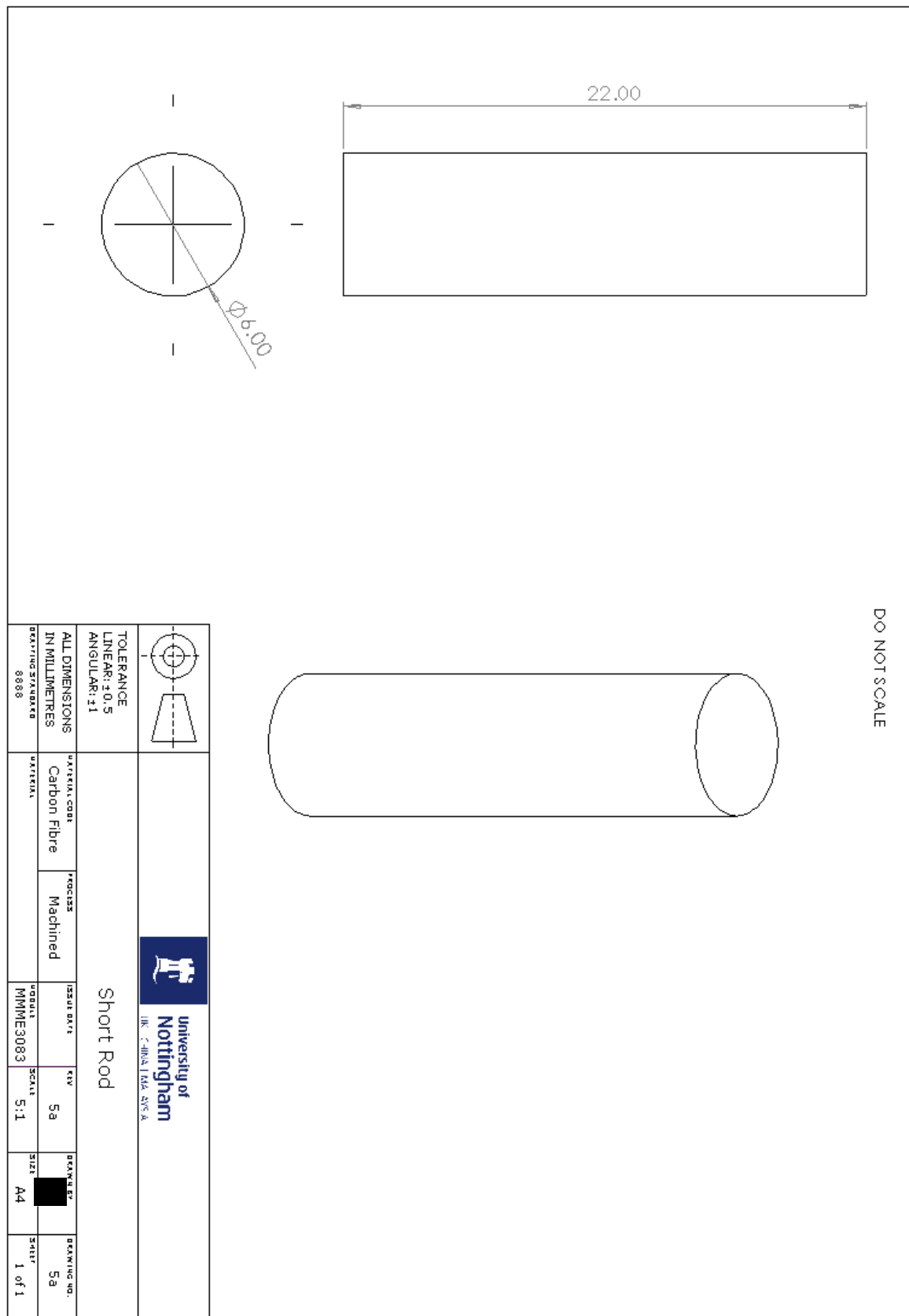


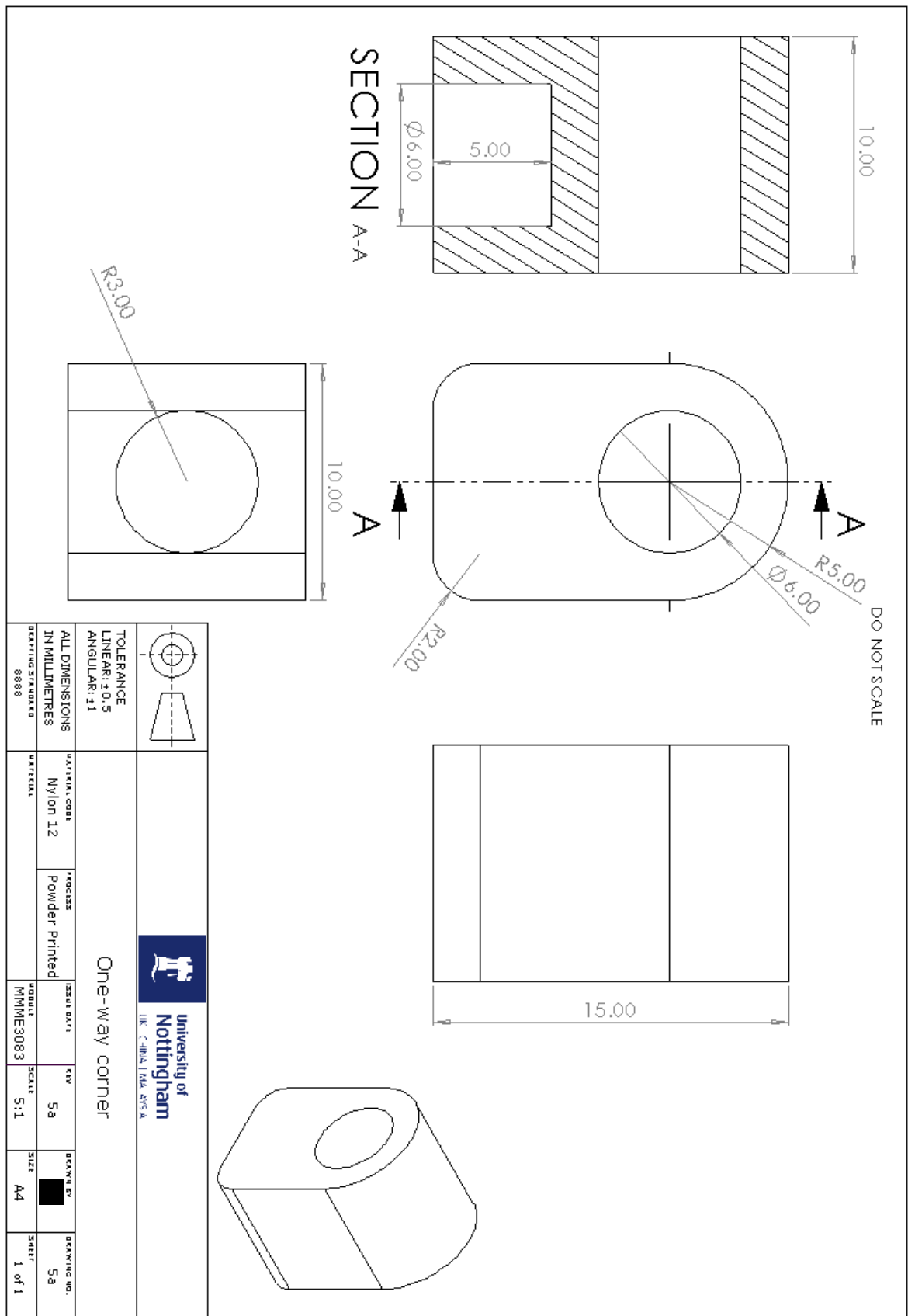


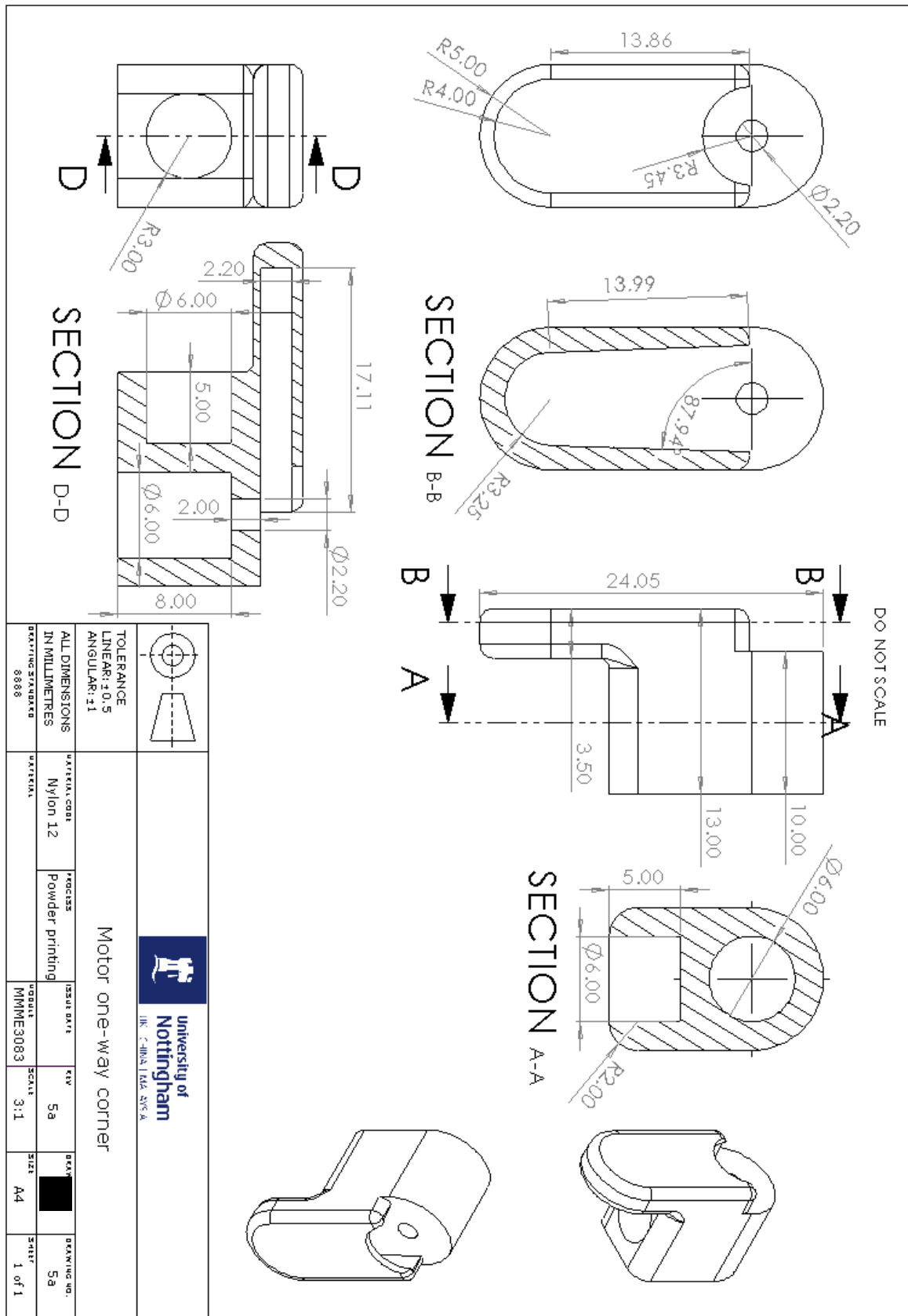


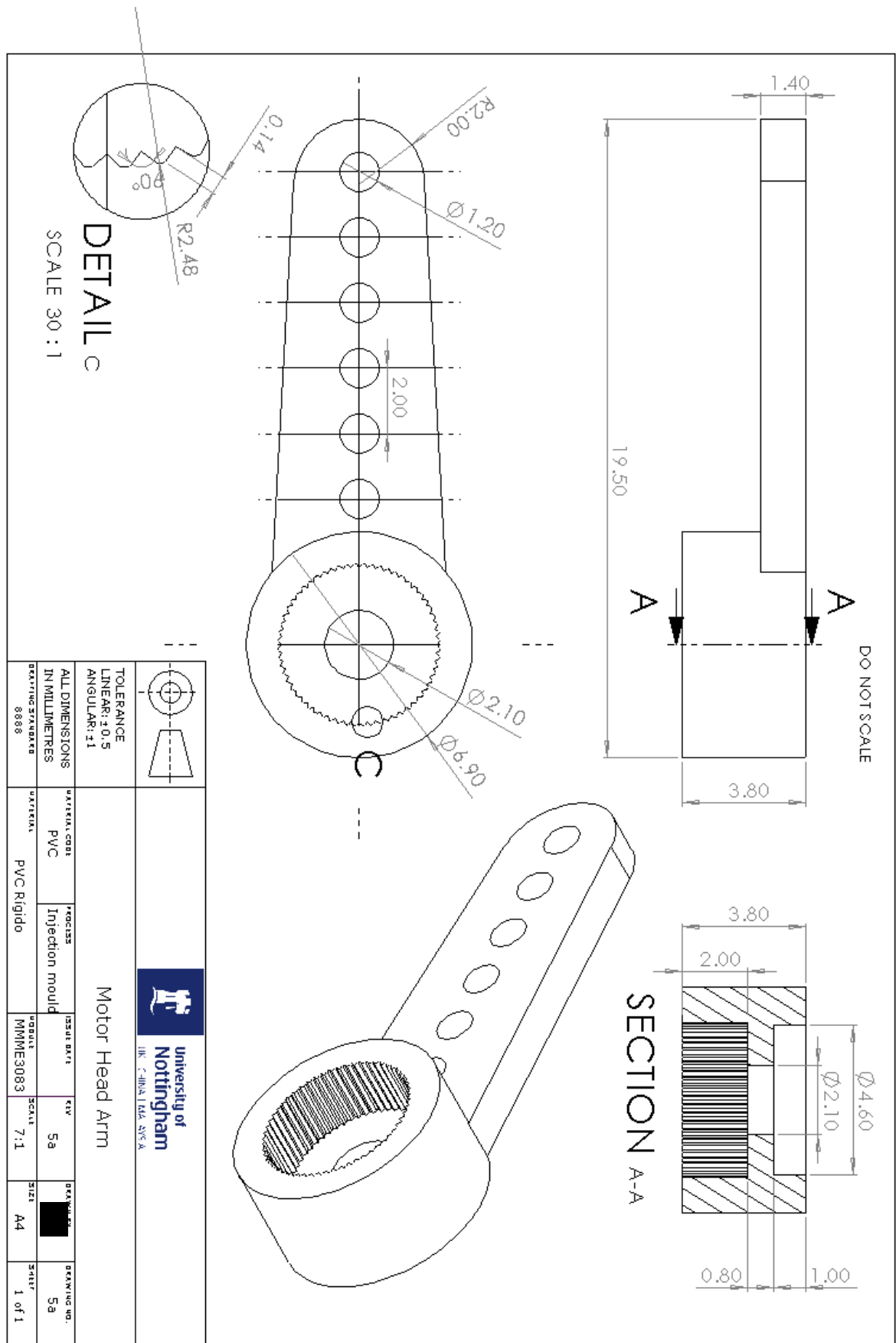












APPENDIX 2 – Sample Codes used in Motor Control Test

Done in Visual Studio Code, in language C.

Test 1: Moving the end effector up and down.

```
#!/usr/bin/python3
import RPi.GPIO as GPIO
import pigpio
import time

servo = 18
servo2=15
servo3=14

pwm = pigpio.pi()
pwm.set_mode(servo, pigpio.OUTPUT)
pwm.set_mode(servo2, pigpio.OUTPUT)
pwm.set_mode(servo3, pigpio.OUTPUT)

pwm.set_PWM_frequency( servo, 50 )
pwm.set_PWM_frequency( servo2, 50 )
pwm.set_PWM_frequency( servo3, 50 )

val = 1000
up = True
step = 5
upper = 1500
lower = 1000
sleep = 0.05

try:
while True:
pwm.set_servo_pulsewidth( servo, val )
pwm.set_servo_pulsewidth( servo2, val )
pwm.set_servo_pulsewidth( servo3, val )
time.sleep(sleep)
if up:
val = val + step
if val > upper:
up=False
else:
val = val - step
if val <lower:
val=lower
up=True
print( "val=", val )
except KeyboardInterrupt:
print("Program stopped")
```

```
# turning off servo
pwm.set_PWM_dutycycle( servo, 0 )
pwm.set_PWM_dutycycle( servo2, 0 )
pwm.set_PWM_dutycycle( servo3, 0 )
pwm.set_PWM_frequency( servo, 0 )
pwm.set_PWM_frequency(servo2,0)
pwm.set_PWM_frequency(servo3,0)
```

Test 2: Moving the end effector side to side.

```
import RPi.GPIO as GPIO
import pigpio
import time

servo = 18
servo2=15
servo3=14

pwm = pigpio.pi()
pwm.set_mode(servo, pigpio.OUTPUT)
pwm.set_mode(servo2, pigpio.OUTPUT)
pwm.set_mode(servo3, pigpio.OUTPUT)

pwm.set_PWM_frequency( servo, 50 )
pwm.set_PWM_frequency( servo2, 50 )
pwm.set_PWM_frequency( servo3, 50 )

val = 0
up = True
step = 10
upper = 250
lower = -250
sleep = 0.05

initial1=1200
initial2=1200
initial3=1200

try:
    while True:
        pwm.set_servo_pulsewidth( servo, initial1 + val )
        pwm.set_servo_pulsewidth( servo2, initial2 - val/2 )
        pwm.set_servo_pulsewidth( servo3, initial3 - val/2 )
        time.sleep(sleep)
        if up:
            val = val + step
            if val >= upper:
                up=False
        else:
            val = val - step
            if val <=lower:
                val=lower
                up=True
        print( "val=", val )
except KeyboardInterrupt:
    print("Program stopped")
```

```
# turning off servo
pwm.set_PWM_dutycycle( servo, 0 )
pwm.set_PWM_dutycycle( servo2, 0 )
pwm.set_PWM_dutycycle( servo3, 0 )
pwm.set_PWM_frequency( servo, 0 )
pwm.set_PWM_frequency(servo2,0)
pwm.set_PWM_frequency(servo3,0)
```